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Late Mesozoic tectonics of Central Asia based on paleomagnetic evidence

Dmitry V. Metelkin [a,b,*], Valery A. Vernikovsky [a,b], Alexey Yu. Kazansky [a,b], Michael T.D. Wingate [c]

[a] Institute of Petroleum Geology and Geophysics, Siberian Branch, Russian Academy of Sciences, Akad. Koptyug ave., 3 Novosibirsk, 630090 Russia
[b] Novosibirsk State University, Pirogova st., 2, Novosibirsk, 630090 Russia
[c] Department of Applied Geology, Curtin University of Technology, Bentley, WA 6163, Australia

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ABSTRACT

This paper presents paleomagnetic data for Late Mesozoic (Middle Jurassic to end-Cretaceous) rocks of the Siberian platform (Verkhoyansk Trough) and its southwestern margin (Transbaikalian basins and Minusa Trough). We determine a series of key paleomagnetic poles for 165, 155, 135, 120, and 75 Ma, which define the Mesozoic apparent polar wander path (APWP) for Siberia. This quantitative approach provides the opportunity for a general revision of Mesozoic tectonics of Central Asia. Many researchers have considered the Eurasian continent to have been completely stable during the Mesozoic era. However, we demonstrate systematic deviations of corresponding Mesozoic poles from Siberia and Europe, and interpret the discrepancies as evidence for large-scale sinistral strike-slip motion due to clockwise rotation of the Siberian plate relative to the European plate. We conclude that, following its Late Paleozoic assembly, the Eurasian plate was not internally stable, i.e. not rigid. The Mesozoic geological evolution of Siberia was dominated by strike-slip tectonics. Rift-related grabens formed within the basement of the West Siberia sedimentary basin and orogenic events occurred along the southwestern margin of the Siberian craton, within the Central Asia tectonic province. Our paleomagnetic reconstruction indicates also that the Mongol-Okhotsk Ocean was still not closed completely before the end of the Jurassic. We propose that final collision occurred in the Early Cretaceous, and during the Middle to Late Jurassic interval, northward subduction of oceanic lithosphere resulted in oblique, west-to-east ocean closure (a "scissors-like" model). The closure was controlled by significant sinistral strike-slip motion of the Siberian craton. This process is reflected in Transbaikalia by extensive bimodal volcanic activity and development of rift-related structures, including pull-apart basins.
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1. Introduction

The continent of Eurasia comprises several major blocks that contain Precambrian basement (Baltica, Siberia, Tarim, N. China, S. China), separated by multiple orogenic belts of various Phanerozoic ages generally referred to as Central Asia (Fig. 1). The main features of Central Asia were formed by the end of the Paleozoic. The closure of ancient oceans and subsequent collisions welded together the main tectonic blocks of Baltica, the Kazakhstan superterrane, and Siberia, during the final episode of the evolution of the Central Asia orogenic belt (e.g. Zonenshain et al., 1990; Şengör and Natal'in, 1996; Lawver et al., 2002; Buslov et al., 2002; Xiao et al., 2003; Buslov et al., 2004; Golonka et al., 2006; Kröner et al., 2007; Xiao et al., 2009; Xiao and Kusky, 2009 and papers therein; Safonova et al., 2009). Many of the old terranes within Central Asia, the West Siberia province for example, are now covered by substantial, almost un-deformed Mesozoic and Cenozoic sedimentary rocks (Fig. 1).

Many authors have considered Eurasia to have been completely stable during the Mesozoic to Cenozoic interval. The paucity of paleomagnetic data allows the assumption of relative rigidity of some large fragments within the belt and an intact northern Eurasia during Mesozoic time. Based on this assumption, a synthetic APW path was constructed for Eurasia as a single rigid block during the Mesozoic and Cenozoic (Besse and Courtillot, 1991, 2002; Schettino and Scotese, 2005). However, it is already clear that large-scale (particularly strike-slip) deformation within the Central Asia orogenic belt produced highly fragmented and contorted structures during Mesozoic time (Bazhenov and Mossakovsky, 1986; Cogné et al., 1999; Bazhenov et al., 1999; Natal'in and Sengör, 2005; Van der Voo et al., 2006; Gilder et al., 2008).

Apart for the Early Triassic, the Mesozoic of Siberia is poorly investigated paleomagnetically in terms of tectonic problems. For Mesozoic time, the Siberian part of the Eurasian plate can be considered as three large tectonic provinces (Fig. 1): West Siberia, Central Asia, and Mongol-Okhotsk (Berzin et al., 1994). Intraplate rifting resulted in formation of the largest sedimentary basin in the West Siberian tectonic province (Surkov et al., 1997). To the south and southeast of Siberia, subduction and subsequent collision between the Siberian and Mongolia-China blocks caused crustal deformation

* Corresponding author. Institute of Petroleum Geology and Geophysics, Siberian Branch, Russian Academy of Sciences, Akad. Koptyug ave., 3 Novosibirsk, 630090 Russia. Tel.: +7 383 335 6433; fax: +7 383 330 9853.
E-mail address: MetelkinDV@ipgg.nsc.ru (D.V. Metelkin).

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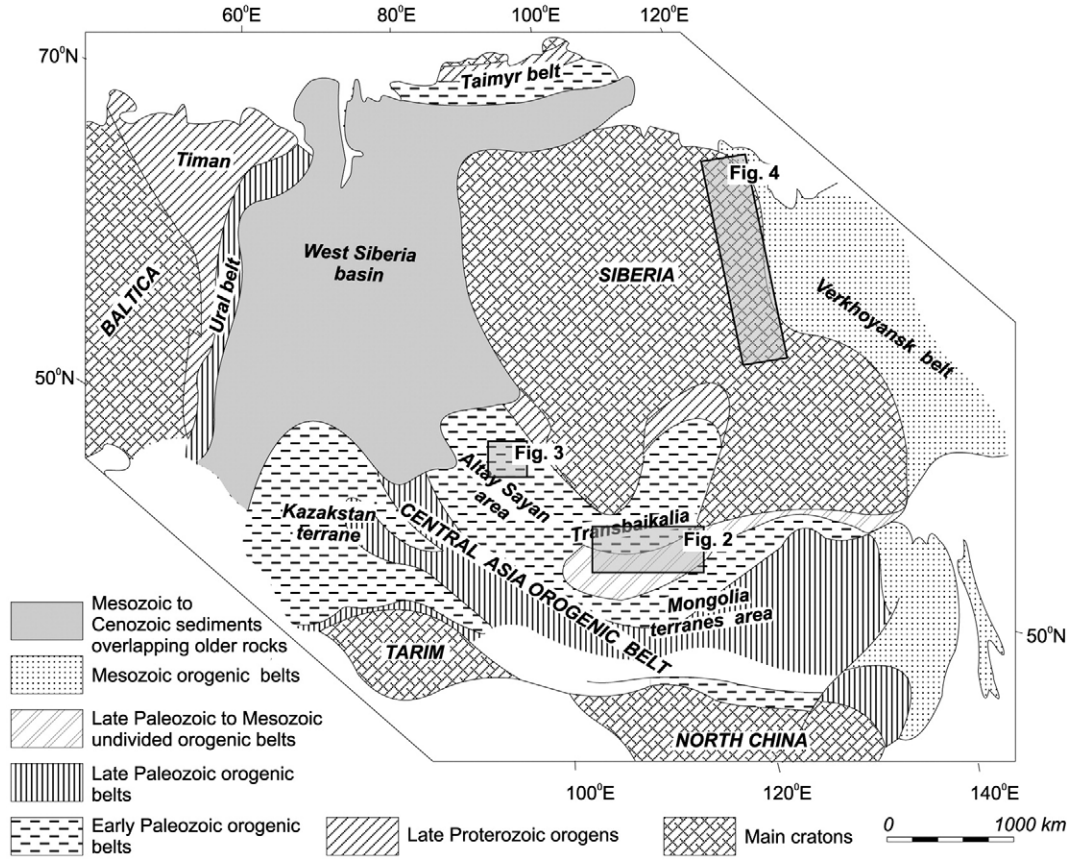


Fig. 1. Main tectonic units of northern Asia (adapted from Berzin et al., 1994). Rectangles indicate studied areas (see Figs. 2–4).

within the Mongol-Okhotsk tectonic province (Zorin, 1999; Gordienko and Kuz'min, 1999; Kravchinsky et al., 2002a,b; Tomurtogoo et al., 2005; Cogné et al., 2005; Golonka et al., 2006). At the same time, the tectonics of the Central Asia province, including the southwestern margins of Siberia, are typical of intraplate environments and intracratonic orogeny (De Grave et al., 2007). During Late Mesozoic time, the internal parts of Siberia formed a typical cratonic platform, with sedimentation occurring mainly on its eastern margin, within the Viluy and Verkhoyansk basins.

This study combines paleomagnetic results for Jurassic and Cretaceous rocks from several tectonic provinces of Siberia. We will first briefly review our results (Metelkin et al., 2004, 2007a,b, 2008)

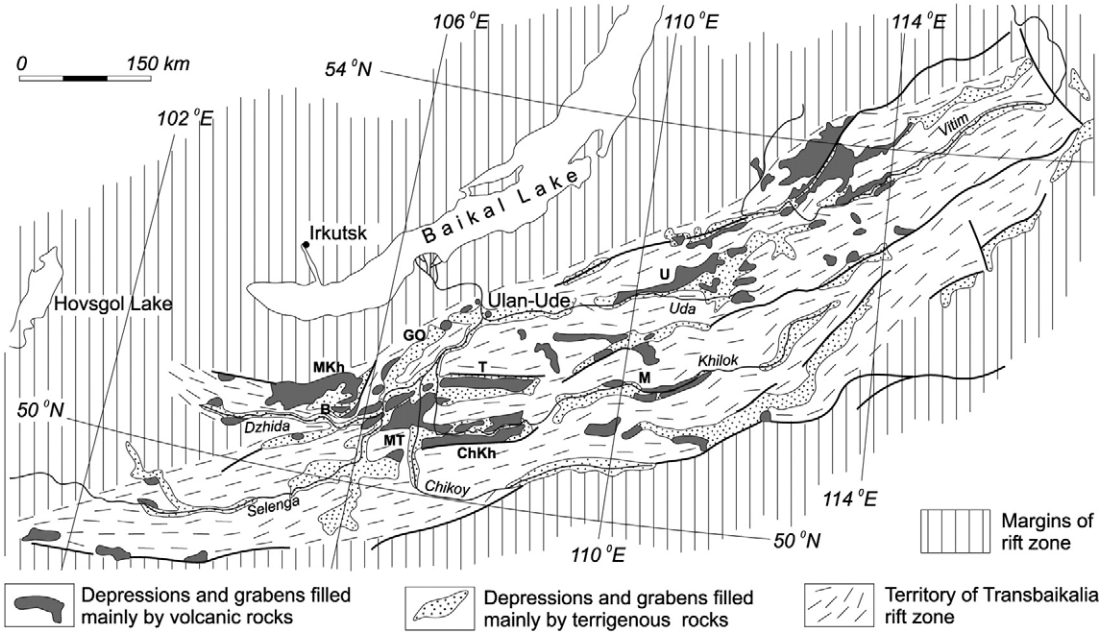


Fig. 2. Schematic structure of the Late Mesozoic to Cenozoic rift zone of Transbaikalia (after Yarmolyuk et al., 1998). Comments: Depressions (volcanic province) lettered as follows: B - Borgoy, ChKh - Chikoy-Khilok, GO - Gusinoe Ozero, M - Mogzon, MKh - Maly Khamar-Daban, MT - Margintuy, T - Tugnuy and U - Uda.

for obvious revision and present a detailed paleomagnetic analysis that constrains the first apparent polar wander path for Siberia for Late Mesozoic time (200-75 Ma). This quantitative approach provides the opportunity to reconstruct intraplate deformation and to generally revise the Mesozoic tectonics of Central Asia.

2. Regional geology and sampling

The studied rocks are located within three different marginal regions of the Siberian platform: Transbaikalia (southern margin), the Minusa Trough (southwestern margin), and the Verkhoyansk Trough (eastern margin). The Verkhoyansk Trough is underlain by Paleozoic sedimentary rocks of the Siberian platform, whereas the Transbaikalia and Minusa Troughs are underlain by Paleozoic basement rocks of the Central Asia mobile belt, which welded together to form the major ancient cratonic blocks (Baltica and Siberia) of Eurasia (Fig. 1). Abundant geological information demonstrates that the accreted blocks of the Central Asia mobile belt were joined to Siberia and can be considered a single tectonic domain since the end of the Paleozoic (e.g. Zonenshain et al., 1990; Berzin et al., 1994; Buslov et al., 2004; Kröner et al., 2007; Xiao et al., 2009). During Mesozoic time, the Minusa and Transbaikalia regions were sites of extensive intraplate magmatic activity (Yarmolyuk et al., 2002; Kovalenko et al., 2004), whereas the Verkhoyansk Trough had a mainly sedimentary environment (Parfenov and Kuz'min, 2001).

2.1. Transbaikalia region

At the southern margin of Siberia, the geology of the Transbaikalia region is dominated by the Late Mesozoic collision between the Mongolia - North China continent and the Siberia - Europe continent as a result of closure of the Mongol-Okhotsk Ocean (Zorin, 1999; Kravchinsky et al., 2002a; Cogné et al., 2005; Tomurtogoo et al., 2005; Golonka et al., 2006; Xiao et al., 2009). Within the Paleozoic accretionary basement, there are numerous large depressions or basins (Fig. 2) associated with strike-slip faults and filled by Mesozoic volcanic and sedimentary rocks (Gordienko and Klimuk, 1995; Gordienko and Kuz'min, 1999). Their structural evolution was accompanied by

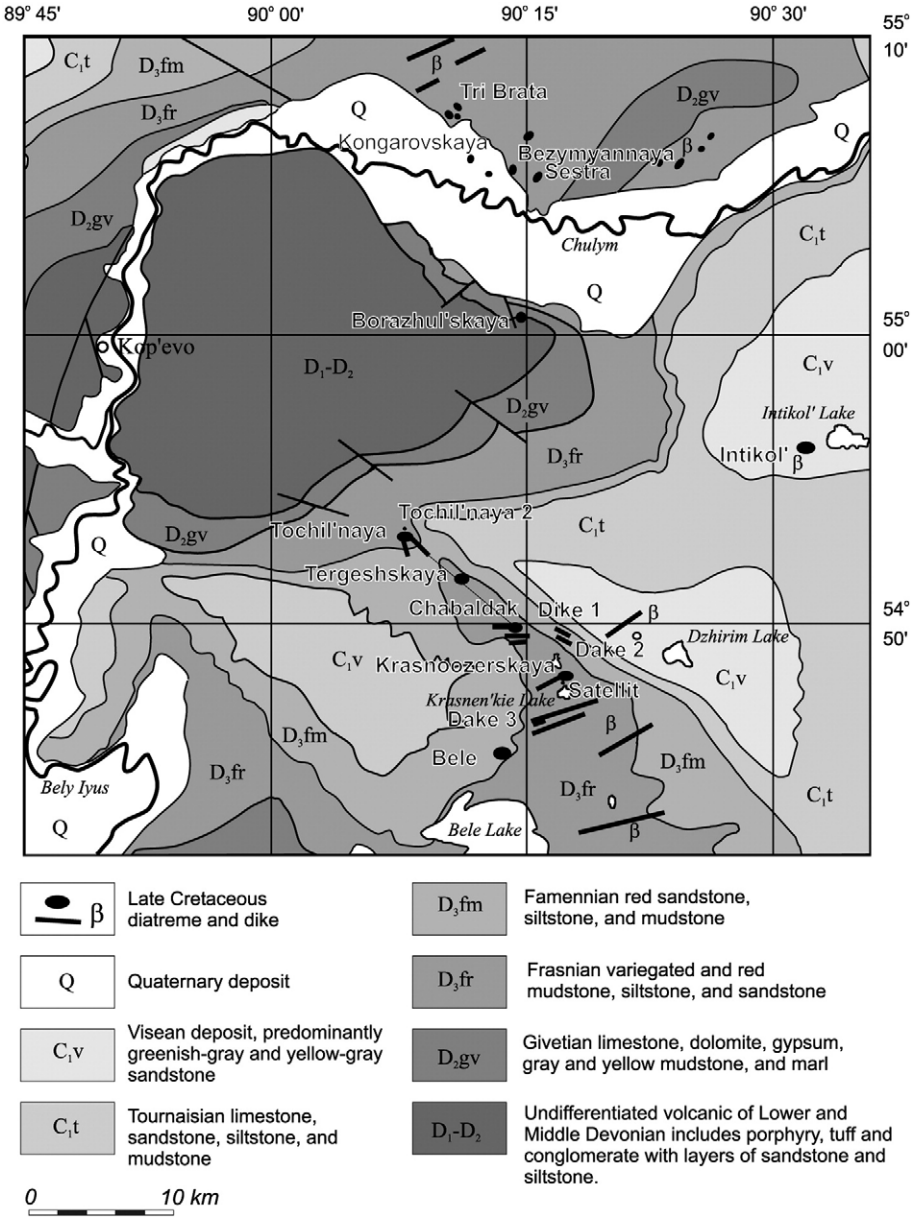


Fig. 3. Geological structure of the Minusa Trough (modified from Metelkin et al., 2007b).

predominantly trachybasaltic volcanism associated with mantle plume tectonics and intracontinental rifting (Gordienko and Kuz'min, 1999). According to Gordienko et al. (2000) this rift-related volcanic activity reflects subduction behind an active margin, similar to modern processes in California. Our paleomagnetic samples were collected from several of the volcano-sedimentary basins shown on Fig. 2 and briefly described below.

2.1.1. Late Jurassic basalts

The Tugnuy volcanic province (T in Fig. 2) is one of the largest Mesozoic intracontinental rift basins in the Transbaikalia region, and resulted from intracontinental strike-slip motion along the Tugnuy-Konda fault zone. Bimodal volcanic and terrigenous sedimentary rocks were deposited from Early Permian to Early Cretaceous time, although the main volcanic activity occurred during the Late Jurassic (Gordienko and Klimuk, 1995) and is represented by the Ithetuy Formation. The Ithetuy Formation consists of subalkaline extrusive units interbedded with conglomerate, sandstone, and siltstone. It has an Upper Jurassic age (from Oxfordian to Kimmeridgian) is based on K-Ar (150 ± 5 Ma) and Rb-Sr (153 ± 2 Ma) isotopic data (Gordienko and Klimuk, 1995; Gordienko et al., 1997). Two lava flows of the Ithetuy Formation, each less than 10 m thick, were sampled in the central part of the Tugnuy basin near Mt. Bytsygyr, and seven horizons, including an intraformational conglomerate layer, were investigated 15 km to the south, along the east bank of the Sukhara river (see Metelkin et al., 2007a for details). In the conglomerate, almost 80% of the clasts are rounded pebbles of volcanic rocks, which together with its structural position, indicates reworking of volcanic material during interruptions in volcanic activity.

Predominantly alkaline and subalkaline trachybasaltic lava fields with several volcanic centres are recognized in the Late Jurassic Margintuy volcanic province (MT in Fig. 2), located between the Dhida, Chikoy, and Khilok Rivers of Transbaikalia (Zhamoitsina, 1997). The Margintuy basalts have a K-Ar age of 156 ± 6 Ma, which is close to the Oxfordian-Kimmeridgian boundary (Gordienko and Zhamoitsina, 1995). In chemical composition and structural position, these rocks are similar to trachybasalts of the Ithetuy formation and are correlated with Late Jurassic bimodal volcanic activity associated with intracontinental rifts along the southern margin of the Siberian platform. We sampled six lava flows for paleomagnetism in a volcanic centre known as Dulan-Khara, close to the village of Kharjasta on the west bank of the Chikoy river (Metelkin et al., 2007a).

The Malo-Chamardaban volcanic province (MCh in Fig. 2) is located to the south of Lake Baikal, along the north bank of the Dhida River and contains Late Jurassic bimodal volcanic rocks of the Ithetuy Formation, including trachybasalt, trachyte, and trachyandesite, with rare thin layers of tuffaceous and terrigenous rocks. The succession is dominated by a series of trachybasaltic lava flows mainly 1.5 to 10 m thick, with some up to 30 m thick. The rarity of the tuffs and abundance of interbedded thick lava flows indicate an intensive eruption process during the Late Jurassic (Litvinovsky et al., 1996). An age of 158 ± 4 Ma for the Ithetuy trachybasalts is based on Rb-Sr and K-Ar isotopic data (Shadaev et al., 1992). Paleomagnetic samples were collected from four 5 to 8 m thick trachybasaltic flows along the east bank of the Armak River in the western part of the Malo-Chamardaban volcanic province (Metelkin et al., 2007a).

2.1.2. Early Cretaceous volcano-sedimentary sections

The Chikoy-Khilok basin, south of Ulan-Ude, between the Chikoy and Khilok Rivers, contains Early Cretaceous rocks of the Chilok Formation (ChKh in Fig. 2). Ar-Ar geochronology indicates that most of the volcanic activity occurred between 122 and 113 Ma (Gordienko et al., 1999). A series of 15 m thick basaltic lava flows, separated by sandstone and conglomerate beds, is exposed along the basin margins (Gordienko et al., 1999). Paleomagnetic samples were collected from four shoshonite lava flows, a sandstone horizon, and an intraformational conglomerate, along the east bank of the Chikoy river, close to Beregovaya village on the west side of the basin. Samples were collected from three trachybasaltic flows of the Khilok volcanics in the southern Chikoy-Khilok basin, near Bichura village, and five flows were sampled on its northeast side near Maleta and Potanino villages, where they are interlayered with subhorizontal sedimentary rocks (see (Metelkin et al., 2004 for details).

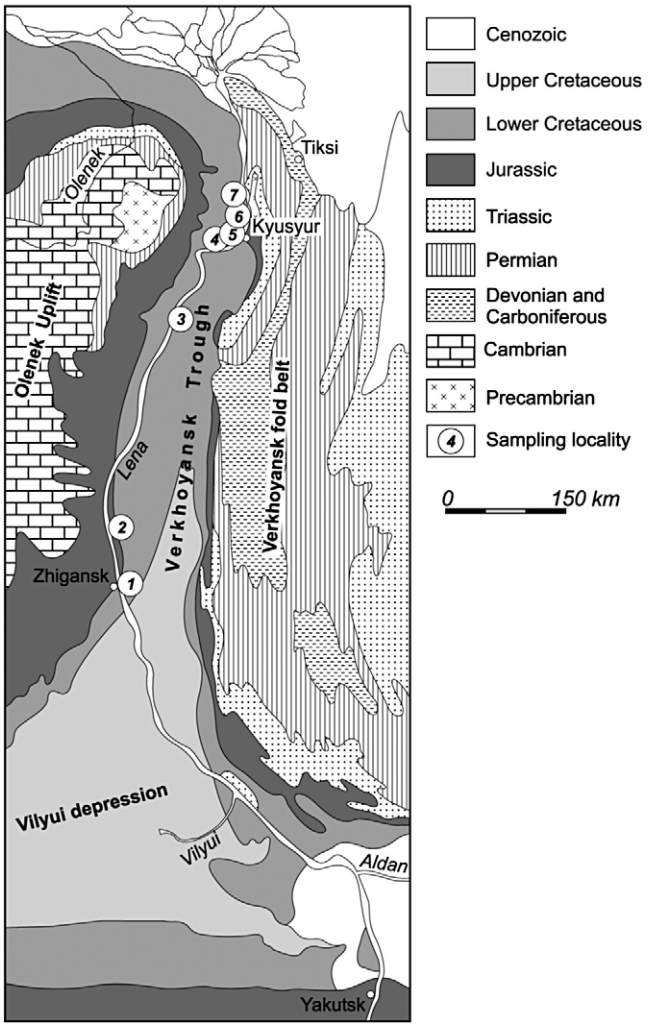
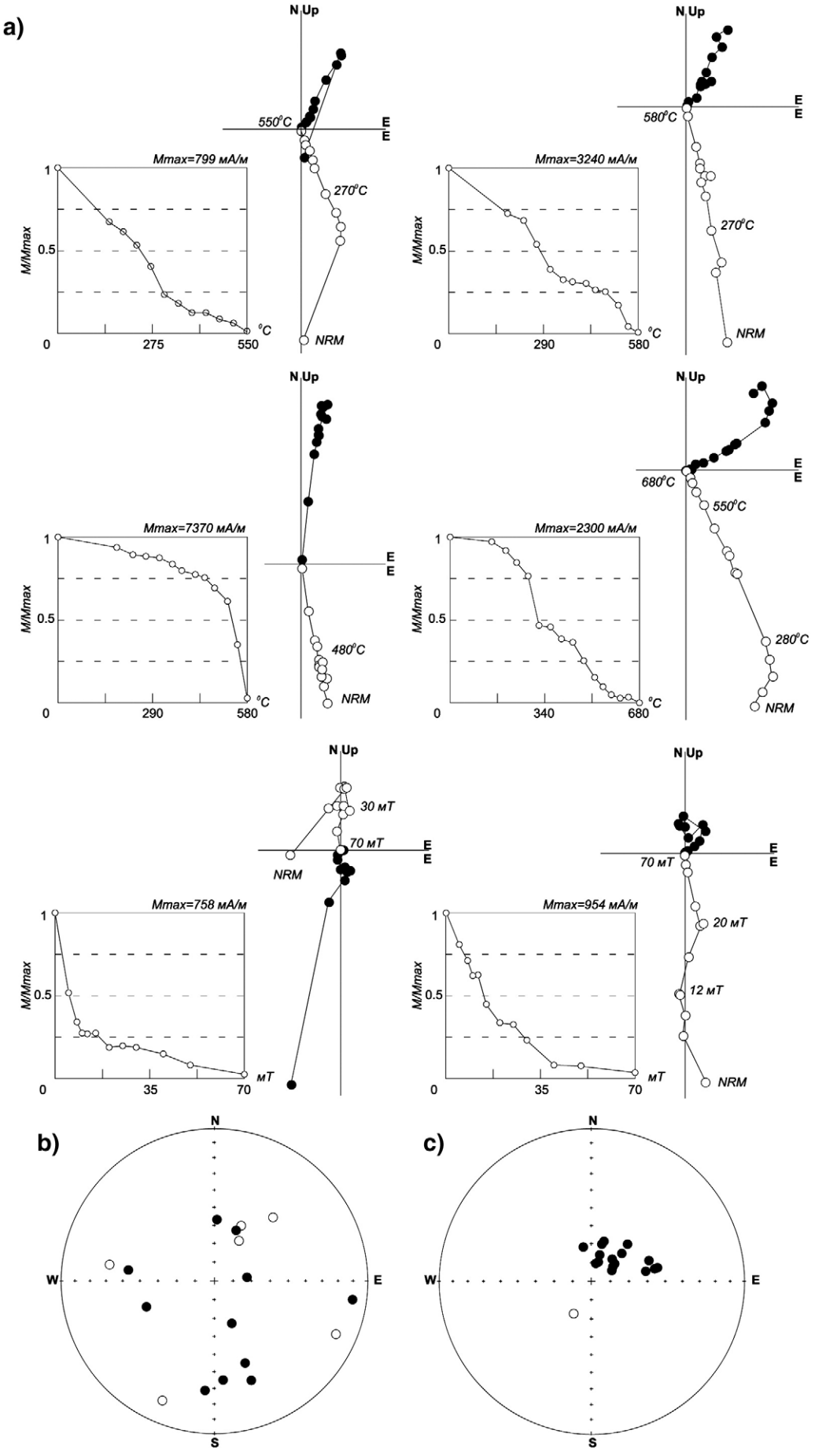


Fig. 4. Geological sketch map of the Verkhoyansk Trough, on the eastern margin of the Siberian platform (modified from Metelkin et al., 2008). Circles denote sampling localities: (1) Zhigansk Town, (2) Cape Kystatym, (3) Cape Obukh, (4) mouth of Kazarma river, (5) Kyusyur Village, (6) Cape Chucha, (7) Cape Chekurov.

About 60 km west of the Chikoy-Khilok and Tugnuy basins, an Early Cretaceous volcano-sedimentary succession in the Borgoy basin (B in Fig. 2) is correlated with the Khilok Formation. The Borgoy basin contains thin basaltic and trachybasaltic lava flows, interbedded with sediments and intrusions (mainly laccoliths, dikes, and sills). K-Ar geochronology suggests that volcanism was active in two stages, with prolonged hiatuses: an initial episode of basaltic volcanism at 136-132 Ma, followed by thin trachybasaltic-teschenitic lava flows and subvolcanic intrusions at 110-102 Ma (Vorontsov et al., 1997). Four individual trachybasaltic lava flows (4 sites) and the two largest teschenite intrusions (3 sites) were sampled for paleomagnetic study (Metelkin et al., 2004).

Early Cretaceous volcanic rocks of the Uda volcanic province (U in Fig. 2), located 150 km east of Ulan-Ude, occur in a series of small depressions along the north bank of the Uda river. Most sections are dominated by thin trachybasaltic lava flows although some contain mainly sedimentary rocks with rare lava flows or sills. Geochemistry


a)
N Up
E
E
550°C
270°C
NRM
Mmax=799 мА/м
M/Mmax
1
0.5
0
275
550
°C
Mmax=3240 мА/м
580°C
290
580
Mmax=7370 мА/м
480°C
Mmax=2300 мА/м
680°C
550°C
280°C
340
680
30 мТ
70 мТ
Mmax=758 мА/м
мТ
35
70
Mmax=954 мА/м
20 мТ
12 мТ
b)
c)
N
W
E
S

and K-Ar geochronology indicate that the trachybasalts formed about 131-126 Ma (Ivanov et al., 1995) and can be correlated with the Khilok formation. Five lava flows (5 sites) were sampled for paleomagnetic study from the area between Bulym and Amgalanta villages (Metelkin et al., 2004).

2.2. Minusa Trough

The Minusa Trough (Fig. 3) contains Late Paleozoic (mainly Devonian) volcano-sedimentary strata that overlie Early Paleozoic rocks of the southwest accretionary margin of Siberia (Berzin and Kungurtsev, 1996). Late Mesozoic intraplate volcanic activity is represented by volcanic necks and dikes of alkaline basalts containing abundant xenoliths of mantle material (Golovin et al., 2000; Mal'kovets, 2001). The presence of xenoliths and the explosive nature of the main eruptions is suggestive of diatremes. Ar-Ar geochronology indicates that this event was of very short duration at 77 ± 5 Ma (Bragin et al., 1999; Mal'kovets, 2001). The diatremes are widespread but clustered in groups of several individual extrusions typically accompanied by small dolerite dikes (Fig. 3). Paleomagnetic samples (more than 200) were collected of basalts from thirteen diatremes and from three associated dikes (see (Metelkin et al., 2007b) for details). Anisotropy of magnetic susceptibility indicates that the bodies have not been tilted since their eruption (Bragin et al., 1999). Owing to the explosive nature of the eruptions, contacts are typically brecciated and poorly exposed, making it difficult to perform baked-contact tests. However, we were able to sample the baked contact of a dike located 4 km to the north of the Krasnozoyrskaya diatreme (Fig. 3).

2.3. Verkhoyansk Trough

The Verkhoyansk foredeep (Fig. 4) is a sedimentary basin situated between the eastern margin of the Siberian platform and the western edge of the Mesozoic Verkhoyansk-Kolyma fold belt. The trough is subparallel to the Lena River, from its lower reaches to the Aldan river, and is filled by Jurassic to Cretaceous fine-grained, coal-bearing molasse, which overlaps thick Paleozoic and Early Mesozoic sedimentary cover of the Siberian platform. The molasse succession includes mainly sandstone interbedded with siltstone, mudstone, and coal-bearing horizons. We studied nine units (formations) of the Verkhoyansk succession at seven localities along the west bank of the Lena River (Fig. 4). Existing detailed stratigraphic data allows definition of the ages of the studied successions to the stratigraphic stage level (Parfenov and Kuz'min, 2001). Deformation of the Late Mesozoic strata infilling the trough reflects the youngest stage of collision events in eastern Siberia, linked to overthrusting of the Verkhoyansk fold belt up to the end of the Cretaceous (Parfenov et al., 1995). This deformation resulted in several large flat-topped structures (e.g. the Chekurovka anticline, where we sampled the main part of the Late Mesozoic section) with steep attitudes (up to 40° and more) which are ideal for the application of paleomagnetic fold tests.

2.3.1. Middle Jurassic terrigenous rocks

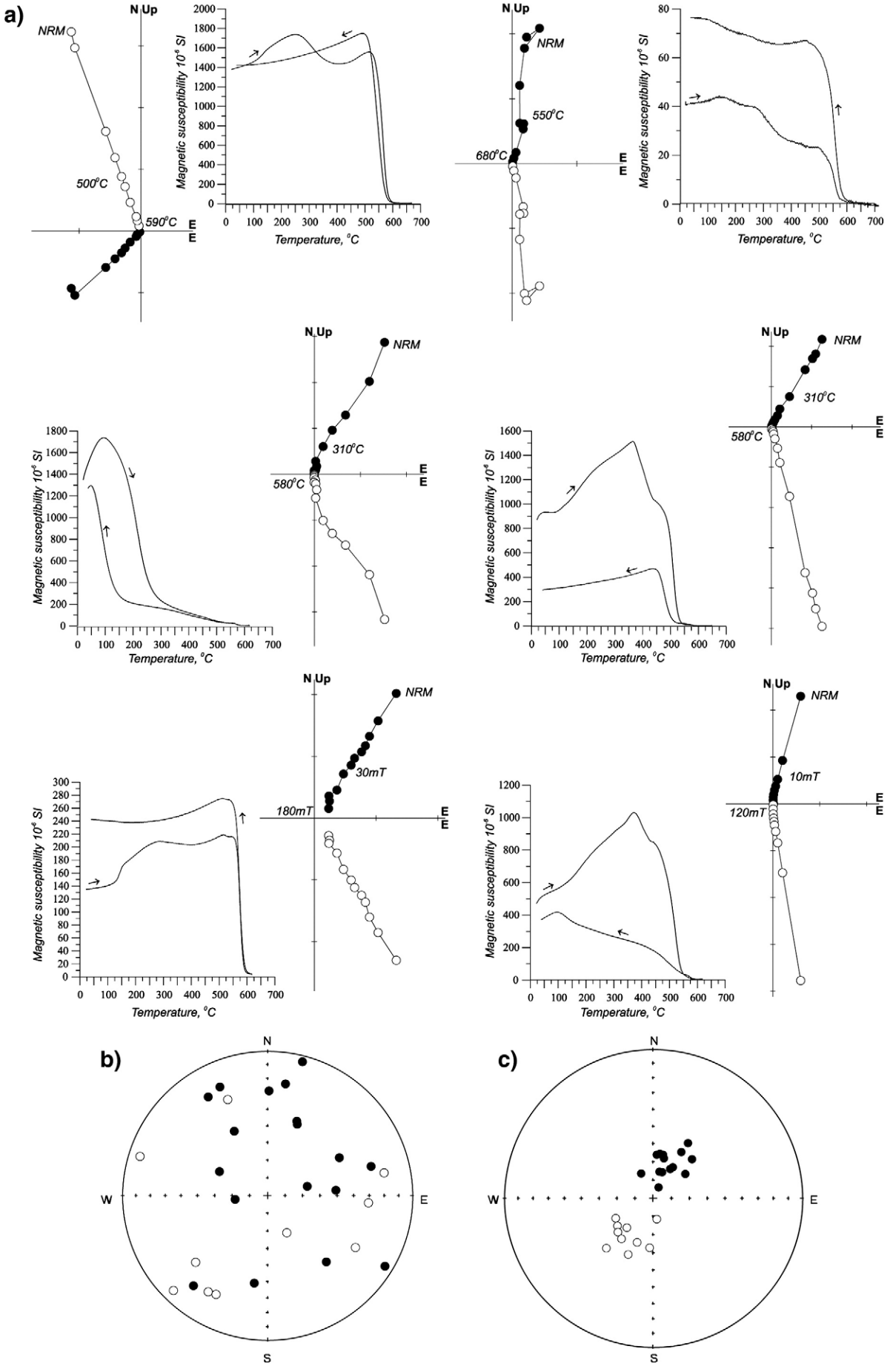
Middle Jurassic strata were sampled at Cape Kystatym (locality 2), located about 75 km north of the town of Zhigansk town, and at Cape Chekurov (locality 7), in the lower reaches of the Lena River (Fig. 4). The Kystatym section includes layers of orange-yellow and grey, fine-grained sandstone of the upper Kystatym Formation. This formation as a whole consists of alternating grey, fine-grained sandstone, grey siltstone, and mudstone containing numerous ammonites (Kirina et al., 1978). The abundant ammonite fauna allows the age of the Kystatym Formation strata to be constrained within the Bathonian stage, between 168 and 165 Ma. At the Cape Chekurov locality, we sampled an anticline with steep limbs within the Chekurov Formation, which consists mainly of thinly layered sandstone, siltstone, and mudstone, with ammonite and pelecypod fossils (Kirina et al., 1978). Two sandstone horizons that correspond (Kirina et al., 1978) to the Bathonian-Callovian stage (163 - 166 Ma) were sampled. In total, from the four sites at two localities, 40 samples of the Middle Jurassic succession were collected for determination of the pole position of Siberia at about 165 Ma.

Table 1
Paleomagnetic directions from Late Jurassic (~155 Ma) basalts of Transbaikalia (after Metelkin et al., 2007a).

Section, lithology, sampling site	n	In situ		Tilt corrected		k	α_{95} (°)
		D (°)	I (°)	D (°)	I (°)		
Tugnuy VTS							
Sukhara, trachybasalt, 04s03	9	39.3	-22.4	43.5	62.3	36.0	8.7
Sukhara, trachybasalt, 04s04	8	30.2	-17.9	28.7	65.7	30.4	10.2
Sukhara, trachybasalt, 04s05	6	22.8	59.9	54.6	76.2	29.2	12.6
Sukhara, trachydolerite, 04s06	15	357.7	60.5	12.5	80.0	201.4	2.7
Sukhara, trachydolerite, 04s08	10	16.0	47.9	60.9	77.8	191.4	3.5
Sukhara, trachybasalt, 04s09	10	34.1	34.1	42.3	73.9	191.4	3.5
Butsygyr, trachydolerite, 04s10	7	191.2	-84.9	208.9	-70.2	29.5	11.3
Butsygyr, trachydolerite, 04s11	11	12.9	59.1	17.9	79.0	72.4	5.4
Mean	8 sites; 76 samples	24.6	40.9	36.9	73.6	4.4 109.3	29.7 5.3
Margintuy VTS							
Dulan-Khara, basalt, 00s14	6	55.1	64.7	52.1	74.6	129.9	5.9
Dulan-Khara, basalt, 00s15	6	33.5	66.9	16.9	75.2	110.5	6.4
Dulan-Khara, basalt, 00s16	8	29.1	59.5	17.0	67.6	266.2	3.4
Dulan-Khara, basalt, 00s17	5	28.8	61.5	15.3	69.4	163.5	6.0
Dulan-Khara, basalt, 00s18	6	10.1	66.9	346.0	71.1	204.1	4.7
Dulan-Khara, basalt, 00s19	5	51.8	58.5	46.8	68.1	92.4	8.0
Mean	6 sites; 36 samples	35.4	63.8	22.1	72.2	97.5 98.4	6.8 6.8
Maly Khamar Daban VTS							
Armak, basalt, 01s10b	11	58.6	63.2	78.0	53.7	50.3	6.5
Armak, basalt, 01s10c	13	57.7	64.8	78.5	55.4	113.9	3.9
Armak, basalt, 01s10d	8	45.8	64.5	69.9	57.3	75.8	6.4
Armak, basalt, 01s10e	12	53.0	69.7	79.4	60.5	37.3	7.2
Mean	4 sites; 44 samples	53.9	65.6	76.5	56.8	460.0 462.3	4.3 4.3

Notes: n - number of samples used in statistics; *D* - paleomagnetic declination; *I* - paleomagnetic inclination, *k* - Fisher (1953) precision parameter; α_{95},- radius of the cone of 95% confidence about mean direction.

Fig. 5. Results from Late Jurassic basalts of Transbaikalia (after Metelkin et al., 2007a). (a) Orthogonal diagrams and associated NRM decay plots illustrating stepwise thermal and AF demagnetizations. (b) High temperature component (HTC) directions from pebbles from the Sukhara intraformational conglomerate. (c) Site-mean directions of stable ChRM after tilt correction (Table 1). In orthogonal diagrams, closed (open) circles represent vector endpoints projected onto the horizontal (vertical) plane. In stereoplots, closed (open) symbols are projected onto the lower (upper) hemisphere.


a)
N Up
NRM
500°C
590°C
E
E
Magnetic susceptibility 10⁻⁶ SI
Temperature, °C
680°C
550°C
310°C
580°C
30mT
180mT
10mT
120mT
b)
c)
N
W
E
S

2.3.2. Early Cretaceous sedimentary rocks

We collected a total of 93 samples of fine-grained sandstone from 11 sites (5 to 10 samples per site) to determine an Early Cretaceous paleomagnetic pole (paleopole) for Siberia. The sandstones have a preliminary correlation with the Berriasian-Valanginian boundary (~140 Ma) based on their stratigraphic position. Early Cretaceous sections were studied at four localities (sites 4-7 in Fig. 4) in the lower reaches of the Lena river: near the mouth of Kazarma river, near Kyusyur village, at Cape Chucha, and at Cape Chekurov. Here, a near-complete sequence of Early Cretaceous deposits is exposed for 100 km along the banks of the Lena River (Petrov, 1980).

Sandstones from the Ogoneryuryakh and Nadbulun Formations were sampled on the west bank of the Lena River, 20 km upstream from Kyusyur village in the mouth of Kazarma River. The stratotype section of the Ogoneryuryakh Formation is located here. Among thick layers of cross-bedded sandstone, the Ogoneryuryakh Formation contains mudstone, siltstone, and coal beds. The age of the strata is thought to be Lower Aptian. The Nadbulun Formation was also sampled in the middle reaches of the Lena River near Cape Obukh (locality 3 in Fig. 4), where the section consists of greenish-gray fine–grained sandstone interbedded with grey mudstone.

Gray fine-grained sandstones of the Kyusyur Formation were sampled in the limb of an anticline near Kyusyur village, where it consists of a Hauterivian sequence of grey mudstone, sandstone, coal-bearing siltstone, and coal members. A continuous succession of Late Cretaceous rocks is traced in cliffs on the west bank of the Lena River, 15 km downstream from Kyusyur village. Here we sampled light-grey massive sandstones of the Kigilyakh and Chonkogor Formations, of Valanginian and Hauterivian-Barremian age, respectively.

Berriasian-Valanginian deposits of the Khairgas Formation were sampled near Cape Chekurov and consist of thick, greenish-grey, massive, fine-grained sandstone strata with rare mudstone layers. A second Berriasian-Valanginian succession, in the Ygnyr Formation, was sampled in the middle reaches of the Lena River near Zhigansk (locality 1 in Fig. 4), where it is composed of interbedded greenish-grey, fine- and medium-grained sandstone, dark-colored mudstone, and coal-bearing horizons.

3. Paleomagnetic methods

Samples were subjected to stepwise thermal or alternating magnetic field (AF) demagnetization in the paleomagnetic laboratories of the Institute of Petroleum Geology and Geophysics SB RAS, Novosibirsk, Russia and the University of California at Santa Cruz (UCSC), USA. In Novosibirsk, samples were heated and cooled in a nonmagnetic oven placed within a triple μ-metal shield (developed by V.P. Aparin, Russia). In the center of the oven, the residual field is about 8 nT. Measurements of remanent magnetization were made with a JR-4 spinner magnetometer. About 50% of the studied samples were measured with a *2G Enterprises* cryogenic magnetometer or *JR-5* spinner magnetometer at UCSC. Thermal demagnetization was carried out with a UCSC-built oven with a residual field about 8 nT. AF demagnetization was performed using the automatic degaussing system built in to the 2G Enterprises magnetometer. To facilitate comparison of results, 10% of the samples from the same blocks were demagnetized and measured in both Santa-Cruz and Novosibirsk. The two sets of laboratory results are virtually identical. Changes in the intensity and direction of remanent magnetization vectors during demagnetization experiments were analyzed using orthogonal vector end-point projections (Zijderveld, 1967). Magnetic component directions were identified using principal component analysis (PCA, Kirschvink, 1980). Site-mean directions were determined using Fisher's (1953) statistics. Data were processed using software developed by Enkin (1994).

Table 2
Paleomagnetic directions from Early Cretaceous (~120 Ma) volcanic and sedimentary rocks of Transbaikalia (after Metelkin et al., 2004).

Section, lithology, sampling site	n	In situ		Tilt corrected		k	α_{95} (°)
		D (°)	I (°)	D (°)	I (°)		
Chikoy-Khilok Basin							
Beregovaya, sandstone, 00-26b	10	300	-70.2	204	-55.9	30.4	8.9
Beregovaya, shoshonite, 00-07	10	310	-62.4	218	-61.5	529.7	2.1
Beregovaya, shoshonite, 00-08	8	312	-58.4	226	-62.9	392.1	2.8
Beregovaya, shoshonite, 01-26a	4	314	-56.8	230	-63.7	652	3.6
Beregovaya, shoshonite, 01-26c	5	323	-59.2	222	-68.4	16.8	19.2
Bichura, trachybasalt, 00-09	7	91.7	86.2	44.1	59.6	159.1	4.8
Bichura, trachybasalt, 00-10	6	13.6	81.2	31.9	53.9	35.5	11.4
Bichura, trachybasalt, 00-11	8	331.6	86.1	30.8	60.3	38.8	9
Maleta, trachybasalt, 01-14a	6	349.9	-86.7	170	-78.3	328.7	3.7
Maleta, trachybasalt, 01-14b	8	198.9	-76.7	184	-62.6	92.1	5.8
Potanino, trachybasalt, 01-15a	10	104.7	79.2	32	69.9	63.7	6.1
Potanino, trachybasalt, 01-15b	6	22.7	85.6	4.1	65.9	67.7	8.2
Potanino, trachybasalt, 01-15c	9	169.2	75.8	23.6	83.4	106.9	5
Mean	13 sites; 97 samples	124.1	78.1	30	65.9	26.7 56.1	8.2 5.6
Borgoy Basin							
Lower, trachybasalt, 01s01	10	0.1	54.3	14.5	67.1	57.9	6.4
Lower, trachybasalt, 01x01	8	359.4	52.5	12.3	65.4	67.3	6.8
Upper, trachybasalt, 01s03	10	337	65.8	12	75	171.3	3.7
Upper, trachybasalt, 01x03	5	345.1	56.6	7.9	65.1	76.5	8.8
Dabkhor, teschenite, 01s02	8	9.3	70.1	29.2	71.5	86.2	6
Dabkhor, teschenite, 01x02	7	356.4	72.1	18.3	74.8	217.6	4.1
Guntui, teschenite, 01s04	11	339.7	70.1	333.4	74.7	99.4	4.6
Mean	7 sites; 59 samples	352.8	63.5	10.7	71.1	68.7 138.5	7.3 5.1
Uda Basin							
Amgalanta, trachydolerite, 01x32	9	10.7	73.1	52.1	68	37.6	8.5
Amgalanta, trachybasalt, 01x33	8	203	-57.2	223	-52	213.3	3.8
Ashanga, trachybasalt, 01x34a	6	219.8	-72.3	241.8	-66.6	42.4	10.4
Ashanga, trachybasalt, 01x34b	9	210.7	-69.8	232.4	-65.2	217.2	3.5
Bulym, trachybasalt, 01x35	5	181.8	-60.8	200	-64.1	29.6	14.3
Mean	5 sites; 37 samples	19.8	67.2	45.2	63.9	74.3 76.2	8.8 8.9

Notes as in Table 1.

Fig. 6. Results from Early Cretaceous rocks of Transbaikalia (after Metelkin et al., 2004). (a) Orthogonal diagrams and *k*(T) decay plots showing representative stepwise demagnetization of Early Cretaceous basalts. (b) HTC directions from pebbles from the Beregovaya intraformational conglomerate. (c) site-mean directions of stable ChRM after tilt correction (Table 2). See Fig. 5 for details.

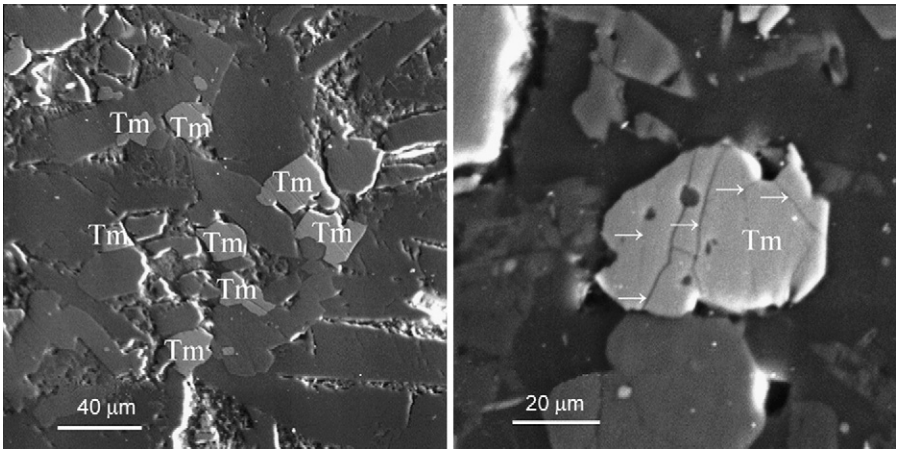


Fig. 7. Photomicrographs of titanomagnetite in basalts of the Minusa diatremes (after Metelkin et al., 2007b). Left image shows homogeneous grains of titanomagnetite (Tm) in basalt of the Chabaldak diatreme. Right image illustrates primary oxidation (marked by arrows) of titanomagnetite in basalts of the Kongarovskaya diatreme.

4. Paleomagnetic results

4.1. Transbaikalia

4.1.1. Late Jurassic basalts

Thermal unblocking temperatures of 550–580 °C (Fig. 5) indicate that magnetite or low-Ti titanomagnetite is the carrier of the high-temperature component (HTC) of the characteristic remanent magnetization (ChRM). A low-temperature component (LTC) is removed after heating up to 250–300 °C or between 300 °C and 350 °C. The LTC is not regular but typically close to the present-day geomagnetic field. These samples are most stable to heating and complete demagnetization is reached at about 680 °C . Nevertheless, the hematite component directions coincide with those indicated by the magnetite ChRM (Metelkin et al., 2007a).

AF-demagnetization was used to isolate the magnetic components in trachydolerites from Mt. Bytsygyr of the Tugnuy volcanic province (Fig. 5). The median demagnetization field (MDF) does not exceed 15 mT, indicating that the NRM is dominated by contributions from low-coercivity minerals. Above 70 mT, more than 90% of initial NRM intensity is removed, and the vector endpoints approach the origin on orthogonal projections. Both normal and reverse paleomagnetic polarities are observed (normal polarity for 04 s10 site and reversed polarity for 04 s11, see Table 1). The reversal test (McFadden and McElhinny, 1990) is statistically significant with classification "C"; the angular distance of $\gamma = 9.2°$ is less than the critical angle of $\gamma_c = 12.3°$ after tilt correction.

Thermal demagnetization of the Sukhara basalts in Tugnuy volcanic province isolates a similar ChRM for six sites (Table 1), see (Metelkin et al., 2007a) for details. A stability test based on clasts within the intraformational conglomerate indicates that the remanence is of primary origin: $R_{obs} = 0.292$, which is less than $R_{crit} = 0.388$, and $k = 1.3$ for $n = 17$. Application of the fold test (McElhinny, 1964) for the site-mean directions from Tugnuy province gives a positive result: $ks/kg = 24.8$ exceeds the critical value (3.7) at the 99% confidence level for $n = 8$. The fold test of Watson and Enkin (1993) indicates an optimum degree of untilting at 112.2%, with 95% confidence limits at 104.6 and 119.6%. The direction–correction tilt test (Enkin, 2003) also gives a positive result ($DC\ Slope = 1.122 \pm 0.142$).

The Fisher (1953) average of site-mean direction for six sites of Dulan-Khara basalts from Margintuy volcanic province does not differ that for the Tugnuy basalts (Table 1). However, due to monoclinal bedding attitudes, a fold test for the Margintuy site-mean directions is inconclusive.

Thermal demagnetization of Armak basalts from the Maly Khamardaban volcanic province isolates a ChRM for four sites, with an average of site-means that is slightly different from the directions from the Tugnuy and Margintuy provinces (Table 1). The fold test for the Armak section is inconclusive, again a reflection of monoclinal bedding. We suggest that the difference between paleomagnetic directions from the Armak basalts and the other provinces reflects: i) local small-scale block rotations during Cenozoic deformation, ii) lack of correction for slope of the volcanic cone (Metelkin et al., 2007a).

However, a synfolding remagnetization is conclusively ruled out by the positive intraformational conglomerate test for the Sukhara section. Moreover, application of a fold test on site-mean directions from all studied Late Jurassic volcanic provinces supports a prefolding origin for the HTC component of the Armak basalts. The site-mean directions group significantly upon tilt correction and yield a positive result for the fold test of McElhinny (1964): $ks/kg = 41.3/8.2 > F_c$ at 99% confidence limit for $n = 18$. The fold test of Watson and Enkin (1993) is also positive, indicating optimum untilting at 95.0%, with 95% confidence limits of 90.2% and 100.4%.

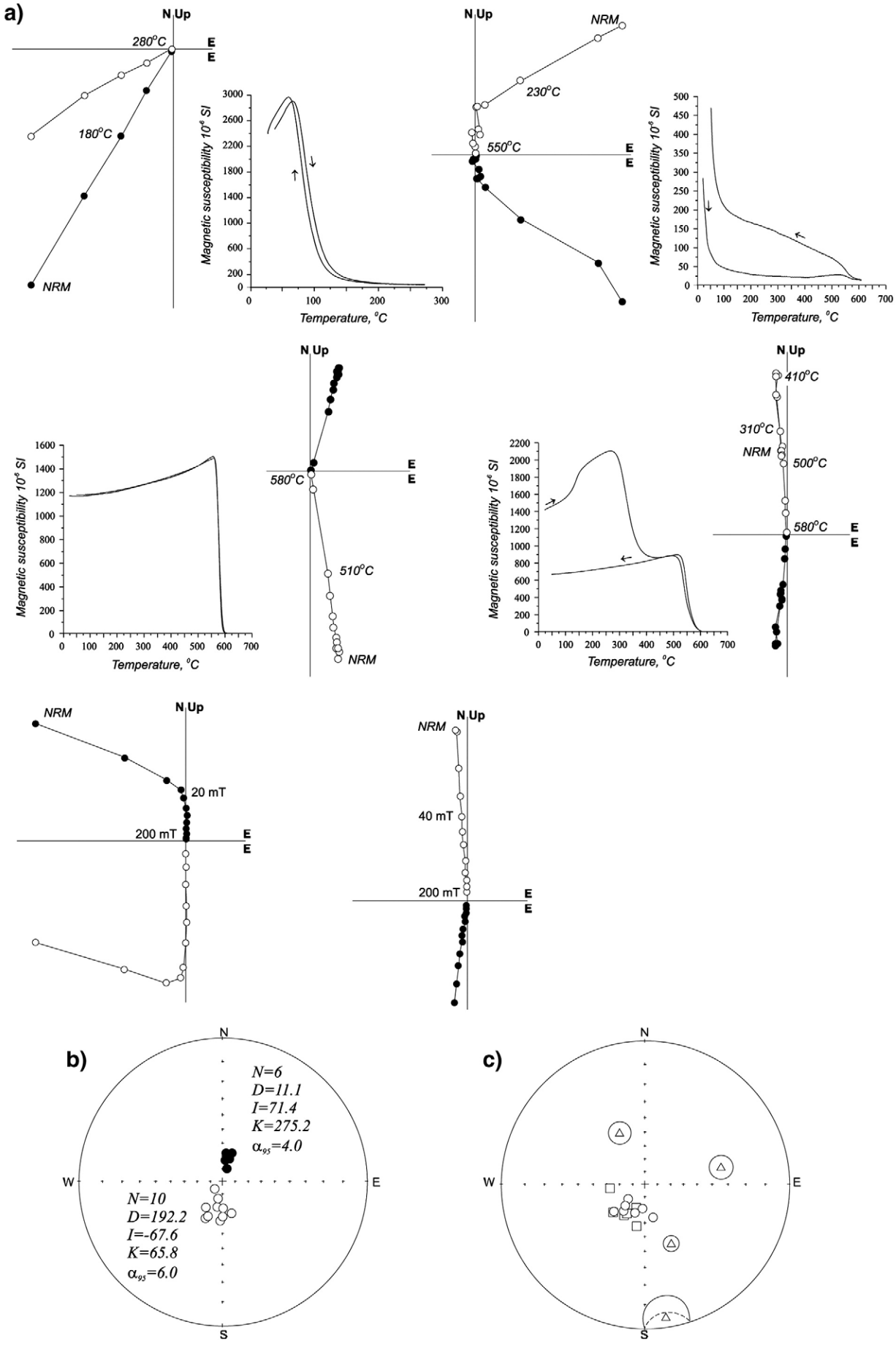
The mean paleopole for the Ichetui Formation, at 63.6°N, 166.8°E ($A_{95} = 8.5$), calculated as the average of virtual geomagnetic poles (VGPs), coincides with the 150-160 Ma pole of Kravchinsky et al. (2002a) for the Badin Formation of the Mogzon basin in Transbaikalia. This good agreement is additional evidence of a Late Jurassic age for the remanence.

4.1.2. Early Cretaceous volcano-sedimentary sections

Unblocking temperatures of 550-590 °C, and magnetic susceptibility range (Fig. 6), indicates that the stable high temperature ChRM is most likely carried by magnetite and perhaps low-Ti titanomagnetite. Hematite is present in some samples, although the remanence direction typically does not depend on magnetic mineralogy and is represented by a single component (Fig. 6). This can be diagnostic of high-temperature oxidation during emplacement of volcanic flows, indicating a primary magnetization acquired during cooling. In some samples, an anomalous low-temperature (<250 °C) component is carried by abundant maghemite.

AF techniques were used to demagnetize about 50% of the samples (Metelkin et al., 2004). In orthogonal diagrams, the majority of magnetization components are directed towards the origin (Fig. 6).

Fig. 8. Results from Late Cretaceous basalts of Minusa province (after Metelkin et al., 2007b). (a) Orthogonal diagrams and k(T) decay plots showing representative stepwise demagnetization. (b) Site-mean directions of stable HTC (Table 3). (c) Baked-contact test showing coincidence of HTC directions from dolerites of Dike 2 (circles) and its baked contact rocks (squares).


a)
N Up
E
E
280°C
180°C
NRM
Magnetic susceptibility 10⁻⁶ SI
Temperature, °C
230°C
550°C
580°C
510°C
410°C
310°C
500°C
20 mT
200 mT
40 mT
b)
N=6
D=11.1
I=71.4
K=275.2
α95=4.0
N=10
D=192.2
I=-67.6
K=65.8
α95=6.0
N
S
W
E
c)

The MDF varies between 10 and 30 mT. Almost all NRM intensity is removed by 120 to 180 mT or more, depending of the proportions of low and high-coercivity minerals. Typically only the ChRM is revealed, although in some samples, both low and high-coercivity components are recognized. Low-coercivity components removed by 30 mT are not consistently directed, whereas the high-coercivity ChRM is well-grouped.

Well-defined ChRM directions of both normal and reverse polarity were isolated from the Chikoy-Khilok basalts by both thermal and AF demagnetization (Table 2). Normal and reversed mean directions differ by 178.0° ($\gamma = 2.0°$, which is less than the critical angle, $\gamma_c = 12.3°$) indicating a positive McFadden and McElhinny (1990) reversal test with "C" classification. The average site-mean direction for Chikoy-Khilok localities based on 13 sites yields a positive McElhinny (1964) fold test at the 95% confidence level ($ks/kg = 2.10 > F_c = 1.98$). The fold test simulation using Watson and Enkin (1993) method indicates optimum untilting close to 100% (82.5% ± 12.7%) implying that the remanence is probably pre-tilting. The direction-correction fold test of Enkin (2003) also yields a positive result (*DC Slope*: 0.825 ± 0.345). The absence of a regional overprint affecting the Beregovaya basalts as well as basalts of the Khilok formation from other localities within the Chikoy-Khilok basin is demonstrated by a positive intraformational conglomerate test. The remanence directions from 28 pebbles of Beregovaya basalts are widely dispersed, with a low precision parameter of $k = 1.2$, and the normalized resultant vector of 0.226 is much less than the critical value of 0.301 (Mardia, 1972).

Both thermal and AF demagnetizations resolve similar ChRM for basalts of Borgoy localities (see (Metelkin et al., 2004) for details). The ChRM direction does not differ significantly between the studied trachybasaltic lava flows and teschenite intrusions from across the Borgoy volcanic province (Table 2). Despite the approximately monoclinal attitude of the lava flows at these localities, and absence of variations in bedding attitudes for teschenites of the Dabkhor and Guntui laccoliths, the site-mean directions show their best grouping in stratigraphic coordinates upon incremental unfolding. Application of the direction-correction tilt test (Enkin, 2003) produces a positive result (*DC Slope*: 1.169 ± 0.898).

Four trachybasaltic flows from Uda volcanic province measured by AF-demagnetization yield similar reversed-polarity ChRMs (Metelkin et al., 2004), whereas one other flow has normal polarity (Table 2). We included nine individual vectors of normal polarity and 28 of reversed polarity to apply a reversal test of McFadden and McElhinny (1990). The angular distance of $\gamma = 6.5$ is less than the critical value of $\gamma_c = 9.7$, implying a positive reversal test with "B" classification. A fold test for the Uda localities remanence is inconclusive, owing to the approximately monoclinal attitude of flows.

In summary, the paleomagnetic data for Early Cretaceous basalts of the Chikoy-Khilok, Borgoy, and Uda volcanic provinces indicates a primary origin for the ChRM. Province-mean directions show a statistically significant increase in grouping upon tilt correction ($k_s/k_g = 116.4/18.2 > F_c$ at 95% confidence level for $n = 3$) and pass the fold test of McElhinny (1964). The fold test of Watson and Enkin (1993) for site-means directions from all provinces indicates optimum untilting close to 100% (90.1% ± 5.7%). The presence of normal (14 sites) and reversed (10 sites) polarity, with a positive reversal test (McFadden and McElhinny, 1990) in classification "B" (the angular distance $\gamma = 6.6$ is less than the critical angle $\gamma_c = 7.7$) after tilt correction. Averaging the individual site VGPs yields a mean paleopole at 72.3°N, 186.4°E ($A_{95} = 6.0$) for Early Cretaceous (~120 Ma) basalts of the Chikoy-Khilok, Borgoy, and Uda volcanic provinces.

Early Cretaceous rocks of Transbaikalia that include sandstones from Gusinoe lake and basalts from Bichura region and Ingoda river localities have also been paleomagnetically studied by Kravchinsky et al. (2002a) and Cogné et al. (2005). Fold test for the paleomagnetic directions of each region is inconclusive and Kravchinsky et al. (2002a) suggested that Early Cretaceous paleopole of Gusinoe lake localities cannot be obtained reliably. Primary magnetizations are supposed despite an inconclusive result of fold test for average directions of Bichura and Ingoda basalts (Cogné et al., 2005). The pole positions for these localities differ significantly from our result as well as among themselves and they are explained by local tectonic rotations as arising from shear movement along the Mongol-Okhotsk suture (Cogné et al., 2005).

Table 3
Paleomagnetic directions from Late Cretaceous intrusions of the Minusa Trough (after Metelkin et al., 2007b).

Diatreme	Age, Ma	n	D (°)	I (°)	k	α_{95} (°)
Tri Brata	75 ± 2.4	8	5.4	72.6	51.4	7.8
Bezymyannaya	—	9	14.8	78.9	41.3	8.1
Intikol'	—	9	6.5	67.2	63.7	6.5
Tergeshskaya	77 ± 1.9	16	15.7	71.0	42.8	5.7
Krasnoozerskaya	77 ± 3.9	30	7.1	72.0	42.1	4.1
Satellit	74 ± 2	19	17.4	66.1	56.6	4.5
Sestra	75 ± 6.2	19	180.1	-62.5	67.9	4.1
Kongarovskaya	74 ± 5.5	29	183.7	-59.4	99.1	2.7
Borazhul'skaya	77 ± 5	20	164.4	-64.4	51.3	4.6
Chabaldak	—	9	178.5	-68.8	53.5	7.1
Bele	79 ± 2	10	203.4	-60.5	42.4	7.5
Tochilnaya	76 ± 1	21	191.6	-69.6	113.2	3.0
Tochilnaya 2	—	13	205.3	-58.5	44.3	6.3
Dike 1	—	17	196.7	-76.8	61.1	4.6
Dike 2	—	13	213.2	-65.9	48.7	6.0
Dolerites		*7*	*208.1*	*-67.6*	*55.1*	*8.2*
Baked contact rocks		*6*	*218.4*	*-63.7*	*40.6*	*10.6*
Dike 3	—	7	228.5	-81.6	159.1	4.8
Mean	~75 Ma	16 sites; 249 samples	16.1	70.1	83.4	4.1

Notes: The $^{40}Ar/^{39}Ar$ ages are from Bragin et al. (1999) and Mal'kovets (2001). Other notes as in Table 1.

4.2. Minusa Trough

Several rock-magnetic techniques (including k(T), Ms(T), and SIRM(T)), together with electron microprobe examinations, were employed to determine the magnetic minerals responsible for the remanence. The first component is related to initial high-Ti titanomagnetites (Fig. 7) with Curie temperatures less than 200 °C (Metelkin et al., 2007b). These grains are mainly multidomain in structure (grain size 10 - 40 μm), and of low magnetic stability. The second NRM component is carried by low-Ti titanomagnetite and/or magnetite that formed during primary oxidation of initial titanomagnetites (Fig. 7), during cooling of basalts below 500 °C. These grains have high magnetic coercivities, Curie temperatures of 500 - 580 °C, and carry the stable high-temperature ChRM (Metelkin et al., 2007b).

The small size and explosive nature of the intrusions (diatremes and dikes) indicates that cooling occurred very quickly, and we consider the $^{39}Ar/^{40}Ar$ age of the intrusions to date the time of remanence acquisition. Primary oxidation had a variable influence on all studied intrusions. In some intrusions (e.g. Tri Brata, Bezymyannaya, Chabaldak, Bele), concentration of initial titanomagnetites is high, and a stable component of NRM is very difficult to isolate (Fig. 8), whereas in others (e.g. Krasnoozerskaya, Kongarovskaya), there is a high concentration of low-Ti titanomagnetites or magnetite and only the well-defined, stable HTC is present (Fig. 8). In some samples, a LTC (directed close to the present-day geomagnetic field) is carried by maghemite, which may have formed in an oxidizing environment long after the primary remanence was acquired.

Thermal and AF demagnetization of basalts (Fig. 8) isolated a stable ChRM from sixteen intrusions (Table 3). Both paleomagnetic polarities are present, with six normal polarity sites and ten sites with

reverse polarity (Table 3). The corresponding polarity-means differ by 176.2° ($\gamma = 3.8°$, which is less than critical angle $\gamma_c = 6.8°$) rendering the reversal test of McFadden and McElhinny (1990) positive with a "B" classification.

The primary origin of the ChRM is also demonstrated by a baked-contact test. The means of remanence from dolerites and baked sandstones do not differ statistically (the angular deviation of 5.8° is less the critical value of 12°), whereas the remanence from Devonian host sediments at some distance from the contact is distinctly different (Kazansky et al., 1996; Metelkin et al., 2007b). Averaging of individual site VGPs yields a mean paleopole at 82.8°N, 188.5°E, ($A_{95} = 6.1$), corresponding to ~75 Ma, and agrees with the 70-80 Ma interval of the European APWP (Besse and Courtillot, 2002; Schettino and Scotese, 2005).

4.3. Verkhoyansk Trough

4.3.1. Middle Jurassic terrigenous rocks

Stepwise thermal demagnetization of Jurassic sandstones from the Chekurov and Kystatym Formations isolated a stable HTC. Representative orthogonal diagrams are shown in Fig. 9. Almost 90 per cent of the total NRM intensity is removed only above 510 °C, and we

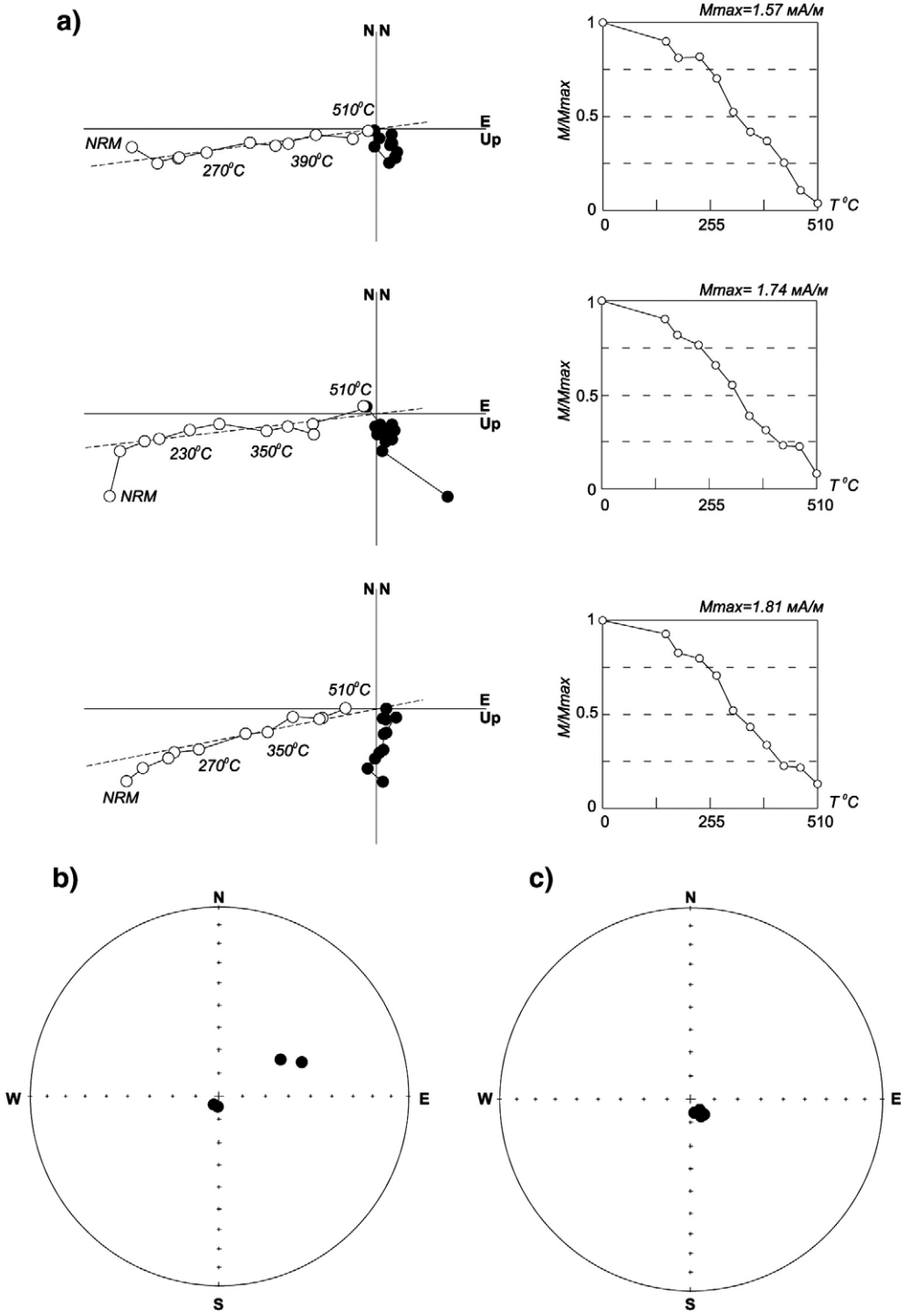


Fig. 9. Results from Middle Jurassic sediments of the Verkhoyansk Trough (a) Orthogonal diagrams and associated NRM decay plots illustrating stepwise thermal demagnetization. Site-mean directions of stable ChRM from Middle Jurassic sediments of the Verkhoyansk Trough (Table 4), both (b) in situ and (c) after tilt correction. See Fig. 5 for details.

Table 4
Paleomagnetic directions from Late Mesozoic rocks of the Verkhoyansk Trough (Lena River region), modified from Metelkin et al. (2008).

Location, sampling site, formation	Age (Ma)	n	In situ		Tilt corrected		k	α_{95} (°)
			D (°)	I (°)	D (°)	I (°)		
Early Cretaceous								
Kazarma River, S4, Ogoneryuryakh Fm	K_1apt (120)	9	333.7	82.0	57.5	76.5	61.8	6.6
Kazarma River, S3, Nadbulun Fm	K_1apt (125)	9	40.5	87.9	67.9	80.2	50.7	7.3
Cape Obukh, S22, Nadbulun Fm	K_1brm (128)	7	74.5	76.7	56.8	62.9	49.1	8.7
Cape Chucha, S20, Chonkogor Fm	K_1hau-brm (130)	10	89.7	74.4	89.1	84.4	127.1	4.3
Cape Chucha, S20, Chonkogor Fm	K_1hau (132)	9	86.8	66.4	118.6	85.6	53.5	7.1
Kyusyur Village, S21, Kyusyr Fm	K_1hau (135)	10	338.9	79.5	65.9	70.3	70.3	5.8
Cape Chucha, S18, Kigilyakh Fm	K_1vlg (136)	10	90.1	62.4	67.7	81.4	57.9	6.4
Cape Chucha, S17, Kigilyakh Fm	K_1vlg (138)	8	81.9	63.6	43.4	80.4	48.9	8.0
Zhigansk Town, S24, Ygnyr Fm	K_1ber-vlg (140)	6	155.8	76.7	111.4	77.3	69.4	8.1
Cape Chekurov, S15, Khairgas Fm	K_1ber-vlg (140)	10	60.4	65.9	98.1	84.3	30.4	8.9
Cape Chekurov, S14, Khairgas Fm	K_1ber (142)	5	57.3	30.3	66.5	80.1	428.4	3.7
Mean	~ 135 Ma	11 sites;	72.1	73.2			17.9	11.1
		94 samples			70.2	79.2	105.6	4.5
Middle Jurassic								
Cape Chekurov, S10, Chekurov Fm.	J_2bt (165)	10	58.9	58.9	129.6	85.3	278.3	2.9
Cape Chekurov, S8, Chekurov Fm.	J_2bt (165)	10	67.3	50.9	163.6	85.3	171.3	3.7
Cape Kystatym, S1, Kystatym Fm.	J_2bt (166)	10	213.3	85.7	132.3	82.6	111.2	4.6
Cape Kystatym, S2, Kystatym Fm.	J_2bj-bt (168)	10	189.3	85.5	144.2	83.0	127.1	4.3
Mean	~165 Ma	4 sites;	69.1	74			13.2	26.3
		40 samples			141.4	84.2	1542.4	2.3

Notes: Age - rock age (in brackets: absolute age in Ma accepted for the pole according to stratigraphic position of the studied rocks). Other notes as in Table 1.

consider magnetite and probably hematite in very low concentration to be the main remanence carriers. In general, sandstones yield only a single component of remanence, although in some samples, an unstable LTC is removed between 150 and 200 °C. All studied sandstones from four sites exhibit a similar HTC (Table 4).

Application of the fold test (McElhinny, 1964) for site-mean directions gives a positive result: $ks/kg = 116.8$ is much higher than the critical value at the 99% confidence level for $n = 4$. The fold test of Watson and Enkin (1993) gives an optimum degree of untilting at 95.6% with 95% confidence limits at 102.1 and 89.6%. The direction–correction tilt test (Enkin, 2003) also gives a positive result: *DC Slope* = 0.958 ± 0.164. The mean paleopole at 59.3°N 139.2°E ($A_{95} = 5.7$), calculated for ~165 Ma, is slightly different from ~155 Ma pole of Transbaikalia.

4.3.2. The Early Cretaceous terrigenous rocks

After heating to 200 °C, a well-defined HTC was isolated from most samples of the Early Cretaceous strata (Fig. 10). The HTC trends usually towards the origin of the orthogonal projections with unblocking temperatures close to 580 °C. Almost 90 per cent of the total NRM intensity is removed already above 510 °C (Fig. 10). The stable ChRM has a steep inclination (70°-85°), typical for Cretaceous rocks of Siberia (Table 4).

The Fisher (1953) average direction of eleven site-means yields a positive fold test (McElhinny, 1964) at the 99% confidence level: $ks/kg = 5.90 > F_c = 2.94$. The fold test of Watson and Enkin (1993) indicates optimum untilting close to 100% (82.8 ± 6.0%), implying that remanence is pre-tilting. The direction-correction fold test by Enkin (2003) also gives a positive result (*DC Slope*: 0.825 ± 0.197). We consider the mean paleopole at 67.2°N, 183.8°E ($A_{95} = 7.8$), calculated by averaging of individual site VGPs, to correspond to ~135 Ma.

4.4. Data summary and Siberian APW path for Late Mesozoic

Results from the Verkhoyansk Trough (Table 4) are based on a near-continuous sedimentary succession, from Bathonian to Aptian, and the succession of VGPs (Fig. 11) primarily reflects polar wander from ~170 to ~120 Ma. Westward wandering of the pole is still apparent after fitting a smooth spline to the data (Fig. 11). The paleopoles from Transbaikalia coincide with the smoothed path, suggesting that the Siberian platform and its folded margins formed a single tectonic domain since at least the Late Jurassic. Based on this assumption, we can employ paleopoles from the Transbaikalia, Minusa, and Verkhoyansk (internal part of the Siberian craton) regions to construct an APWP for Siberia. The key average pole positions used are listed in Table 5. The APWP for Siberia is similar to the European APWP (Fig. 11). Both paths show westward pole drift through the Jurassic and a sharp turn to drift north during the Cretaceous, reflecting the common tectonics of both Eurasian domains during the Mesozoic.

However, there are well-defined differences between the Siberian and European APWPs. The Siberian poles are offset from the European poles, occupying more southerly positions prior to the Late Cretaceous. The angles between corresponding poles decrease gradually from Jurassic to the end of the Cretaceous. These systematic deviations are consistent with large-scale intraplate strike-slip motions between the Siberian and European cratons, resulting from the clockwise rotation of Siberia.

5. Discussion and conclusions

5.1. Tectonics of Central Asia

The mid-Mesozoic paleopoles for Siberia are statistically different from the corresponding poles for both Europe and Mongolia-China. The discrepancy between the expected and observed paleomagnetic directions for Siberia in Table 6 illustrate a general clockwise rotation of Siberia relative to the European and Asian blocks up to the end of Cretaceous. We have calculated the Euler pole which provides the best fit of Siberian poles to the corresponding poles for Europe. In doing this, we follow the criteria proposed by Zonenshain et al. (1990), who posed the problem of Mesozoic rotation of Siberia and was probably the first to estimate the scale of this rotation. The criteria are: 1. the Euler pole must be located within the Siberian craton, and 2. the rotations of Siberia must be clockwise.

The calculated Euler pole, at 60°N, 115°E, agrees well (within 10°) with Euler poles reported previously (Bazhenov and Mossakovsky, 1986; Zonenshain et al., 1990; Voronov, 1997; Kazansky, 2002). The best fit of Jurassic poles for Siberia to the European APWP is achieved via counterclockwise rotations (Fig. 11) of 45° (for 165 Ma) and 28° (for 155 Ma), whereas for Cretaceous poles, the rotations are 17° (for 135 Ma) and 12° (for 120 Ma). The average rate of rotation probably

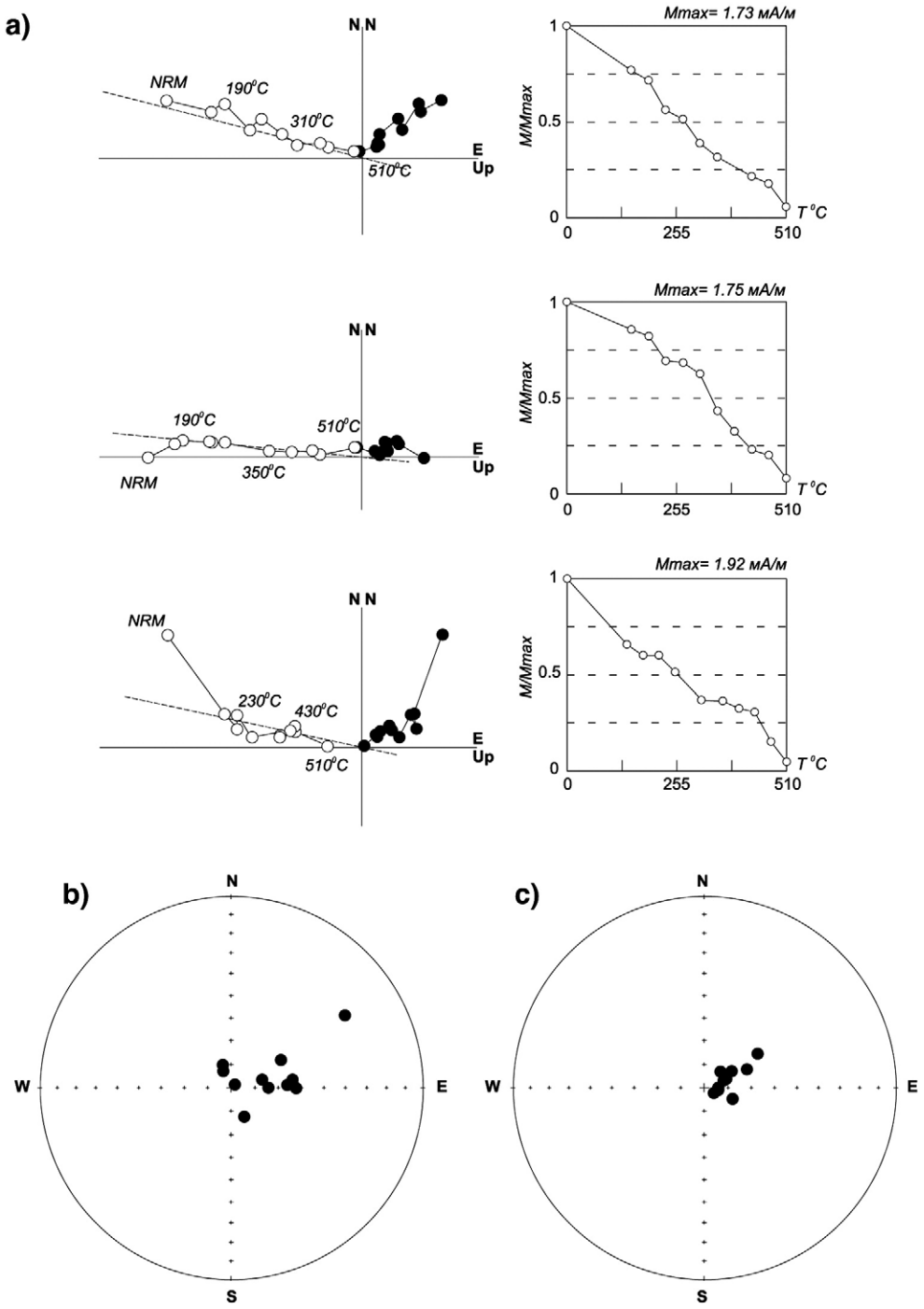


Fig. 10. Results from Early Cretaceous sediments of the Verkhoyansk Trough (a) Orthogonal diagrams and associated NRM decay plots illustrating stepwise thermal demagnetization. (b-c) Site-mean directions of stable ChRM (Table 4), both (b) in situ and (c) after tilt correction. See Fig. 5 for details.

did not exceed 0.7°/Ma (according to the APWP, the maximum rate is 2.5°/Ma at about 160 Ma). The data suggest that Late Mesozoic relative strike-slip displacements were up to 500 km. We conclude, therefore, that after Late Paleozoic assembly, the Eurasian plate was not internally rigid. This contrasts with previous conclusions that the main cratonic blocks of Baltica and Siberia, and the Central Asian fold belt, had already welded together by the Early Permian and moved as a single plate since that time (e.g. Besse and Courtillot, 2002).

Mesozoic reorganization of Central Asia occurred under conditions of intraplate strike-slip deformation, as has been proposed by Bazhenov et al. (1999), Natal'in and Sengör (2005), Van der Voo et al. (2006) and Gilder et al. (2008). We have expanded the model of Natal'in and Sengör (2005), illustrated in Fig. 12, wherein large-scale sinistral wrench-faulting between the main tectonic domains of Siberia, Europe and Kazakhstan, crumpled the Central Asia structure. The main blocks of Eurasia as whole converge within this large-scale sinistral system. According to Xiao et al. (2009), in the Permian the structures of Mongolia and Northern China were separated by the Solonker ocean. It is proposed that the closure of this ocean and formation of Tien Shan - Solonker orogenic belt was propagated between the late Permian and middle Triassic in the west and early/middle Triassic in the east (Xiao et al., 2003, 2009; Chen et al., 2009).

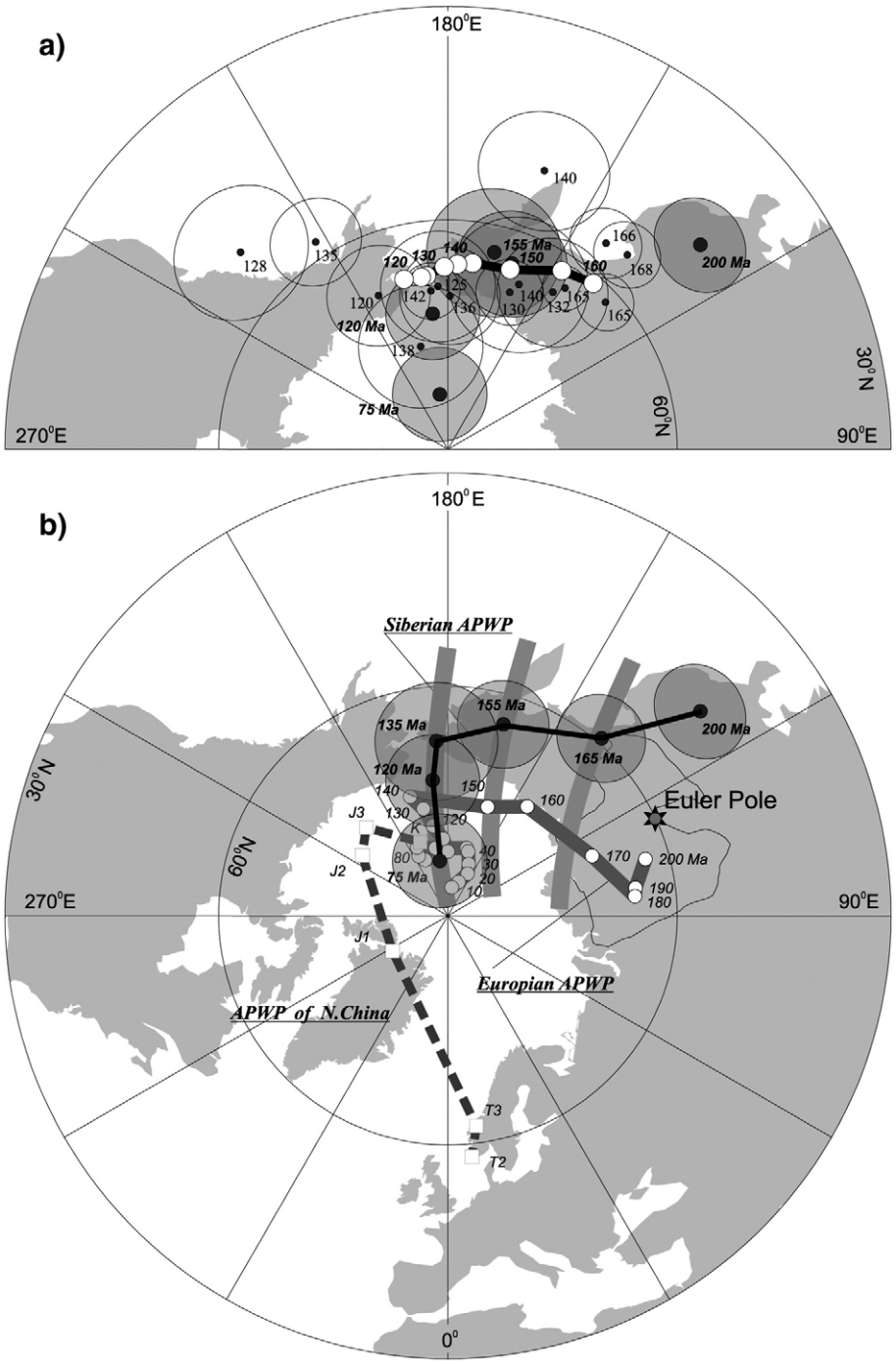


Fig. 11. Mesozoic poles from Siberia and reference blocks of Eurasia (modified from Metelkin et al., 2008). (a) virtual geomagnetic poles, VGPs, (small black circles) from studied Jurassic–Cretaceous sedimentary successions of the Verkhoyansk Trough (Table 4). The black line with large white dots indicates averaged positions of the VGPs using a spline fit and employing a moving average method (calculated for a 20 Myr sliding window every 5 Myr). Larger black circles indicate paleopoles from the Transbaikalia and Minusa regions of the Siberian craton, as well as the generally accepted ~200 Ma pole for Siberia (Table 5). All poles are shown with A_{95} confidence ovals. (b) selected Mesozoic poles (with shaded A_{95} ovals) for Siberia (Siberian APWP) compared with the reference APWPs for Europe (Besse and Courtillot, 2002) and North China (after (Zhao et al., 1996; Gilder and Courtillot, 1997; Yang and Besse, 2001; Uno and Huang, 2003; Lin et al., 2003; Huang et al., 2008). Refer to Table 6 for data details. Thick grey curves represent small circles passing through coeval Siberian and European poles and centered on an Euler Pole at 60°N, 115°E.

Most likely, formation of the Solonker suture occurred against the background of block rotation of Mongolia and China. These conclusions are consistent with the proposed reconstruction. It is obvious that strike-slip tectonics dominated the Mesozoic geological evolution of Central Asia (Fig. 12). We believe that deformation of the Central Asian crust was associated with relative motions of separate components of its composite structure (the Siberian, European, and Kazakhstan tectonic domains) along a system of large sinistral strike-slip zones, in a regime of general clockwise rotation of the Eurasian plate. Deformation of the Asian part of the plate within the limits of the model occurred along a series of shear zones to east (in modern coordinates) closure of the Mongol-Ohotsk ocean which separated the Siberian margin from a collage of Paleozoic terranes of Mongolia and China. Geological implications of such a tectonic scenario in Siberia fit well with constructions stated in Bazhenov and Mossakovsky (1986) and Voronov (1997). Owing to the geometry of the main borders of

Table 5
Key paleopoles for Siberia used for APWP construction.

Rock units	Age range (Ma)	N/n	Test	Paleopole (°)				References
				Lat	Long	A_{95}	T (Ma)	
Late Cretaceous Diatremes of the Minusa Trough	74-82	16/243	Rb, C+	82.8	188.5	6.1	75	Metelkin et al. (2007a)
Early Cretaceous basalts of Transbaikalia (Khilok Fm.)	110-130	25/193	Rb, F+, G*+	72.3	186.4	6.0	120	Metelkin et al. (2004)
Early Cretaceous sediments of the Verkhoyansk Trough	140-120	11/93	F+	67.2	183.8	7.8	135	Metelkin et al. (2008)
Late Jurassic basalts of Transbaikalia (Ichetuy Fm.)	150-160	18/156	Rc, F+, G*+	63.6	166.8	8.5	155	Metelkin et al. (2007b)
Late Jurassic basalts of Transbaikalia (Badin Fm.)	150-160	12/86	Ro, F+	64.4	161.0	7.0	155	Kravchinsky et al. (2002a)
Middle Jurassic sediments of Verkhoyansk Trough	170-160	4/40	F+	59.3	139.2	5.7	165	Metelkin et al. (2008)
Triassic - Early Jurassic Lena River Sediments	175-245	26/26	-	47.0	129.0	9.0	200	*Pisarevsky (1982)

Notes: These averaged poles are used to construct the APWP for Siberia shown in Fig. 11. N/n, number of sites/number of samples; Test, paleomagnetic tests: R - reversal test, where "b" and "c" indexes correspond to (McFadden and McElhinny, 1990) classification, "o" - means test inconclusive; F+ positive fold test; C+ - positive baked-contact test; G*+ - positive intraformational conglomerate test; T - mean age (in Ma) accepted for pole. *Pisarevsky, 1982: pole #4417 from the IAGA Global Paleomagnetic Database (http://www.ngu.no/geodynamics/gpmdb/), that confirm by paleopole for Early Jurassic basalts of Tugnuy province (Monostoy river of Transbaikalia) Lat = 43.3°N, Plong = 131.4°E, A_{95} = 23.0 (Cogné et al., 2005).

the Siberian domain, strike-slip movements, together with clockwise rotation, of the domain should result in a steady compressional environment within the central-Asian province, and an extensional environment in the north of the West-Siberian province. Geological evidence of these processes includes: i) Early Triassic rifting and subsequent dynamics of development of the West-Siberian Mesozoic-Cenozoic sedimentary basin (Surkov et al., 1997), and ii) remobilization of tectonic structure and orogenic events within Altai-Sayan folded area (De Grave et al., 2007; Buslov et al., 2008). The strike-slip movements have a discrete character as is demonstrated in the reconstructed multi-stage history of the main orogenic epoch (De Grave et al., 2007; Buslov et al., 2008) and in the temporal consistency of strike-slip motion and other deformation of the Mesozoic sedimentary complex of Western Siberia (Belyakov et al., 2000). 3-D modelling of seismic data readily illustrates modern deformation structures in pre-Quaternary deposits of the West-Siberian basin,

Table 6
Expected paleomagnetic directions and relative motions of Siberia, based on comparison of mean Mesozoic paleopoles for Siberia, Europe, China, and Mongolia.

Block	Age (Ma)	Pole			References	Siberia expected directions				Tectonic motion	
		Lat	*Long*	A_{95}		*Plat*	*D*	*I*	α_{95}	*F*	*R*
Late Cretaceous (75 Ma)											
SIB	75	82.2	188.5	6.1	Metelkin et al. (2007a)	61.3	15.7	74.4	3.5		
EUR	75	81.3	188.6	7.2	Besse and Courtillot (2002)	61.3	17.6	74.7	4.1	0.0 ± 6.9	1.9 ± 14.4
NCB	K2	81.1	194.0	11	Zhao et al. (1996)	60.5	18.0	74.2	6.5	-0.7 ± 8.1	2.2 ± 19.1
SCB	K2	75.2	210.7	7.5	Lin et al. (2003)	55.5	26.7	71.1	4.6	-5.8 ± 7.1	11.0 ± 13.5
MON	K2	84.7	190.3	6.7	Hankard et al. (2007)	60.9	10.6	74.5	3.9	-0.4 ± 6.7	5.1 ± 13.8
Early Cretaceous (120 Ma)											
SIB	120	72.3	186.4	6.0	Metelkin et al. (2004)	60.9	36.3	74.4	3.5		
EUR	120	78.2	189.4	2.4	Besse and Courtillot (2002)	61.1	24.0	74.6	1.4	0.1 ± 4.8	12.3 ± 9.8
NCB	K1-2	79.8	200.6	4.5	Lin et al. (2003)	59.2	20.2	73.4	2.7	-1.7 ± 5.5	16.1 ± 11.1
SCB	K1	78.8	202.1	7.1	Lin et al. (2003)	58.7	21.9	73.1	4.2	-2.1 ± 6.8	14.4 ± 13.5
MON	K1-2	77.3	215.2	7.5	Hankard et al. (2005)	55.6	22.5	71.1	4.6	-5.2 ± 6.6	13.8 ± 13.3
TAR	K1-2	64.1	172.1	12	Gilder et al. (2003)	63.9	56.3	76.2	6.8	3.3 ± 9.8	20.0 ± 21.9
Early Cretaceous (135 Ma)											
SIB	135	67.2	183.8	7.8	Metelkin et al. (2008)	60.3	46.8	74.1	4.5		
EUR	135	76.8	190.0	5.5	Besse and Courtillot (2002)	60.8	26.9	74.4	3.2	0.4 ± 6.8	19.9 ± 14.2
NCB	K1	81.7	206.8	6.7	Lin et al. (2003)	58.7	16.1	73.1	4.0	-1.6 ± 7.5	30.6 ± 14.9
SCB	K1	78.8	202.1	7.1	Lin et al. (2003)	58.7	21.9	73.1	4.2	-1.6 ± 7.7	24.9 ± 15.3
MON	K1-2	77.3	215.2	7.5	Hankard et al. (2005)	55.6	22.5	71.1	4.6	-4.7 ± 7.5	24.2 ± 15.1
Late Jurassic (155 Ma)											
SIB	155	63.6	166.8	8.5	Metelkin et al. (2007b)	66.0	59.0	77.4	4.7		
SIB	155	64.4	161.0	7.0	Kravchinsky et al. (2002a)	68.6	58.4	78.9	3.8	2.6 ± 8.0	0.6 ± 20.8
EUR	150	75.0	159.9	6.6	Besse and Courtillot (2002)	68.2	29.4	78.7	3.6	2.1 ± 7.9	29.6 ± 20.1
NCB	J3	74.6	220.0	9.4	Huang et al. (2008)	53.2	25.3	69.5	5.9	-12.8 ± 9.3	33.7 ± 19.2
SCB	J3	71.8	224.8	8.6	Yang and Besse (2001)	50.3	27.4	67.5	5.7	-15.7 ± 8.9	31.6 ± 18.2
MON	J3	68.5	231.6	9.5	Zhao et al. (1990)	46.4	28.4	64.5	6.6	-19.7 ± 8.5	30.7 ± 18.3
Middle Jurassic (165 Ma)											
SIB	165	59.3	139.2	5.7	Metelkin et al. (2008)	77.8	82.7	83.8	2.9		
EUR	165	70.6	149.5	9.7	Besse and Courtillot (2002)	72.5	38.7	81.0	5.1	-5.2 ± 8.0	44.0 ± 30.9
NCB	J2	76.4	234.8	5.1	Yang and Besse (2001)	51.6	19.2	68.4	3.3	-26.4 ± 5.6	63.6 ± 20.7
SCB	J1-2	79.9	221.8	5.3	Yang and Besse (2001)	55.8	17.4	71.2	3.2	-22.1 ± 5.7	65.4 ± 21.0
MON	J2-3	68.6	261.8	4.1	Kravchinsky et al. (2002a)	40.8	15.3	59.9	3.1	-37.2 ± 5.1	67.4 ± 20.2
TAR	J2	75.7	198.0	8.1	Gilder et al. (2008)	58.7	28.1	73.1	4.8	-19.2 ± 7.3	54.6 ± 22.9

Notes: Pole - pole position for indicated block (SIB - Siberia; EUR - Europe, NCB - North China Block, SCB - South China Block, MON - Mongolia, TAR - Tarim); Siberia expected directions are paleolatitudes (Plat) and paleomagnetic directions expected from corresponding pole for the point 60°N, 115°E in Siberia; Tectonic motion - displacement of Siberia relative to corresponding block (*F* - poleward displacement (positive northward). *R* - rotation angle (positive clockwise), calculated using PMSGC v.4.1 program (Enkin, 1994). All quantities except Age are in degrees; other notes as in Table 1.

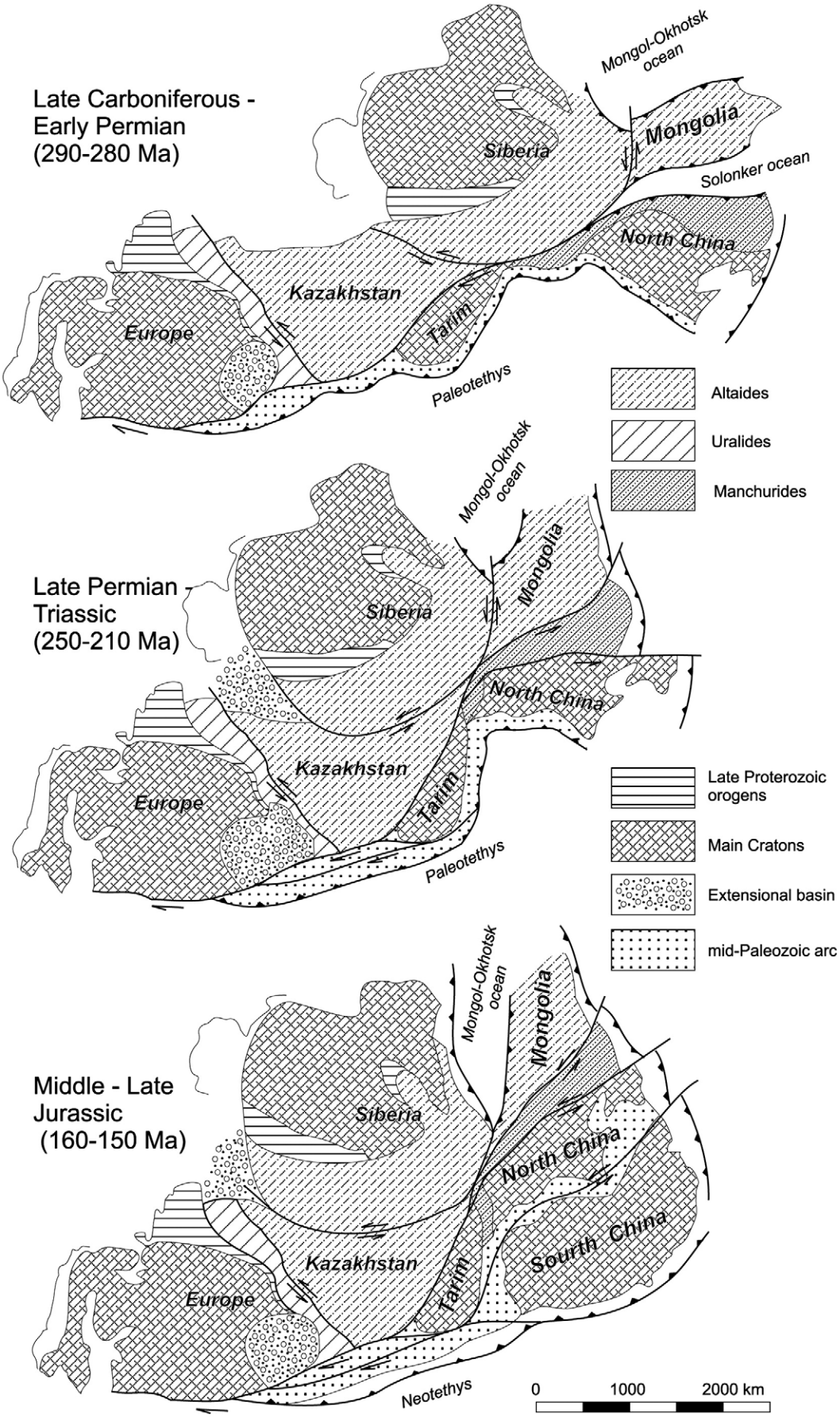


Fig. 12. Late Paleozoic to Early Mesozoic paleotectonic reconstruction of Eurasia shows sinistral transpression between the converging Siberia and Baltica resulted from clockwise rotations and intraplate shifts of Siberian tectonic domain. Adopted and modified from (Natal'in and Sengör, 2005; Van der Voo et al., 2006).

definitely caused by strike-slip faulting in Pre-Mesozoic basement and attributed to graben-rift system of the basin (Koronovsky et al., 2009).

The perfect fit of 75 Ma poles from both Siberia and Europe attests to the termination of strike-slip activity related to clockwise rotation of Siberia up to the end of Cretaceous. However, similar modern intraplate activity in Siberia has been proposed for the Cenozoic and observed in GPS data, although the Cenozoic movements are too small for precise paleomagnetic definition (Timofeev et al., 2008).

A peculiar tectonic environment within the proposed model characterizes the Mongol-Ohotsk province. First of all, the existence of an extensional environment within the Transbaikalian region directly follows from the model (Bazhenov and Mossakovsky, 1986). Systems of rift depressions widespread across the region are attributed to large northeast-striking fault zones (Yarmolyuk et al., 1998; Gordienko and Kuz'min, 1999; Gordienko et al., 2000; Yarmolyuk et al., 2002). More than 200 depressions in the Transbaikalian region are infilled by products of intraplate bimodal magmatism and intercontinental coarse-grained deposits (Yarmolyuk et al., 1998; Gordienko and Kuz'min, 1999; Gordienko et al., 2000; Yarmolyuk et al., 2002; Kovalenko et al., 2004). The depressions formed from the Late Permian to Early Triassic, although the maximum development occurred in the Late Triassic, Jurassic, and Early Cretaceous (Gordienko and Kuz'min, 1999). Sporadic pulses of intercontinental rifting, which continue today in the Baikal Rift system, and the correspondence of extensional structures with strike-slip faults are in close agreement with the model.

5.2. Closure of the Mongol-Okhotsk Ocean

Our paleomagnetic reconstructions constrain the kinematics of the closing of the Mongol-Okhotsk Ocean. Comparisons of Middle and Late Jurassic paleopoles from Siberia with those from North and South China (Gilder and Courtillot, 1997; Yang and Besse, 2001; Huang et al., 2008), Mongolian blocks (including Inner Mongolia (Zhao et al., 1990), and Amuria (Kravchinsky et al., 2002a; Cogné et al., 2005), and Tarim (Gilder et al., 2008) all situated to the south of the Mongol-Okhotsk suture, indicate discrepancies of >10° in inclination and >20° in declination between the expected and observed paleomagnetic directions for Siberia (Table 6). These discrepancies imply significant relative tectonic rotations and poleward displacements (Table 6), and suggest closure of an ocean, from about 3000 km wide at 165 Ma to about 1500 km at 155 Ma, between these tectonic domains.

For example, the 155 Ma poles from Siberia and Mongolia-China require at least 4° (~400 km) of poleward displacement (Table 6). The poleward distance is not exactly the width of the ocean. Although this distance depends also on the relative positions of Siberia and Mongolia-China bloks (rotation of the one block relative to another), the distance between Siberia and Mongolia-China exceeds the uncertainty of the paleomagnetic method. Deformation and crustal thickening as a result of collision following the closure of the Mongol-Okhotsk Ocean could have shortened the distance between the Siberia and Mongolia-China blocks. However, it is more likely that this difference reflects the approximate width of an ocean. Ocean closure can result in significant local rotations of heterogeneous tectonic units within the zone of interaction between continental domains as proposed by Kravchinsky et al. (2002a) and Cogné et al. (2005). However, the kinematic parameters (Table 6) indicate that the regional rotations were controlled by large-scale sinistral strike-slip displacements. Hence we conclude that the convergence of Siberia and Mongolia-China during closure of the Mongol-Okhotsk Ocean was achieved by clockwise rotation of the Siberian block. Note that the angular displacements of Siberia relative to Europe and of Siberia relative to Mongolia-China are similar, because both reflect the same tectonic process.

Paleomagnetic evidence indicates that the Mongol-Okhotsk Ocean was still not completely closed at the end of the Jurassic (Fig. 13). This

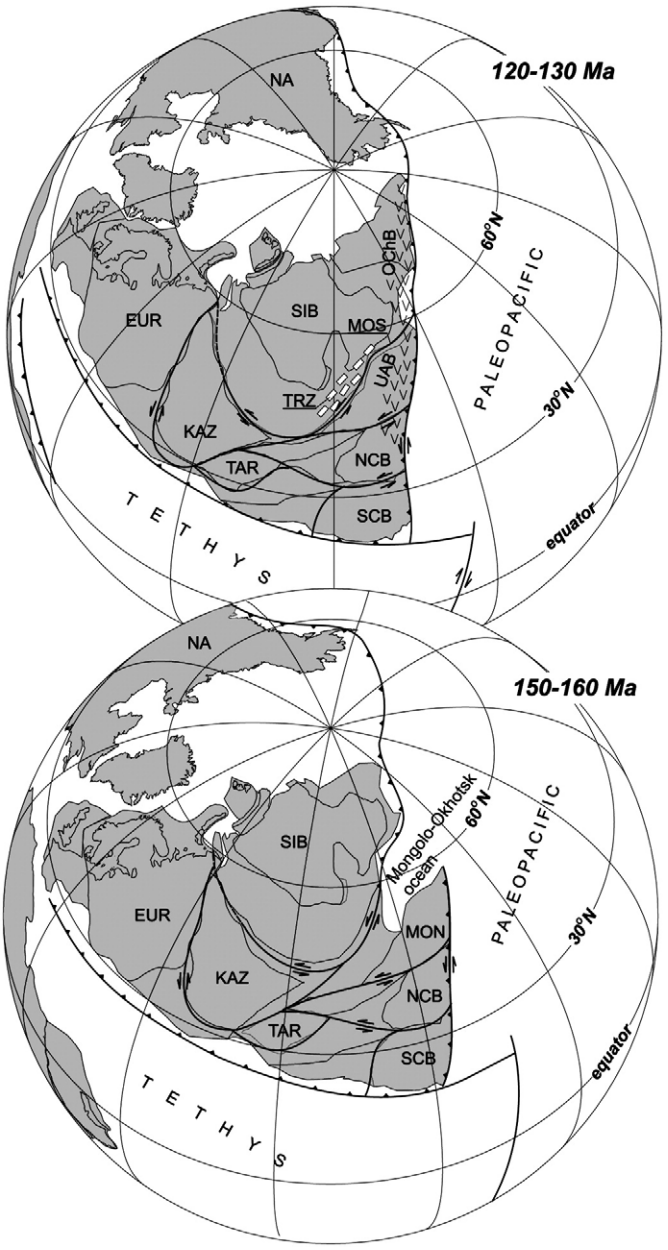


Fig. 13. Schematic reconstruction of the Mongol-Okhotsk ocean closure from Late Jurassic to Early Cretaceous based on paleomagnetic data (see Tables 5 and 6), modified from (Metelkin et al., 2004). SIB – Siberia, EUR – Europe , KAZ – Kazakhstan, NA - North America, NSB – North China, SCB – South China, TAR - Tarim, MON - Mongolia, OChB - Okhotsk - Chukcha volcanic belt, UAB - Upper Amurian volcanic belt, MOS - Mongol-Okhotsk Suture, TRZ – Transbaikalia rift zone. Rectangles schematically show locations of Mesozoic basins in Transbaikalia; 'v' indicates volcanic rocks.

contrasts with an earlier conclusion that in the Middle to Late Jurassic, the Mongol-Okhotsk belt underwent a typical collisional orogeny (e.g. Zorin, 1999; Tomurtogoo et al., 2005). Kravchinsky et al. (2002a) proposed that at 155 Ma there were shallow intracontinental sedimentary basins, which were subsequently shortened, overthrust, and folded in the continuing process of Siberia–China convergence. We propose that final ocean closure and collisional orogeny took place in the Early Cretaceous, and that during the Middle-Late Jurassic interval, ocean closure propagated from west to east (Delvaux et al., 1995), according to a "scissors-like" model, as has been proposed previously (Zhao et al., 1990; Zonenshain et al., 1990; Scotese, 1991; Kravchinsky et al., 2002a; Tomurtogoo et al., 2005; Golonka et al., 2006; Xiao et al., 2009). Thus, the Mongol-Okhotsk Ocean was closed by the Late Jurassic in the west and by the Early Cretaceous in the east

(e.g. Kravchinsky et al., 2002b; Tomurtogoo et al., 2005), and the process was controlled by significant sinistral strike-slip motion of Siberia (Fig. 13). The process is represented in Transbaikalia by intensive bimodal volcanic activity and development of rift-related structures, such as pull-apart basins. The Early Cretaceous paleopole for Siberia almost exactly coincides with the reference poles for both Europe (Besse and Courtillot, 2002; Schettino and Scotese, 2005) and Mongolia-China (Lin et al., 2003; Hankard et al., 2005, 2007), consistent with complete closure of the Mongol-Okhotsk Ocean at this time. However, clockwise rotation of Siberian domain continued at least until the end of Cretaceous time.

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